



Part 1:

DC Analysis

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Getting started

What is Symbolator?

Symbolator (“**S**ymbolic Circuit **S**imulator”) is a small TI-Basic program widely regarded as the best simulator of linear circuit ever made for a calculator. It can perform direct current, alternating current, transient time-domain and complex frequency-domain analyses of numerical and symbolic linear circuits. It can find the Thévenin, Norton and two-port equivalent of a circuit. It accepts elements such as resistors, inductors, mutual inductances, capacitors, independent and dependent current and voltage sources, ideal operational amplifiers, ideal transformers and six types of two-ports. It can also do Bode plots.

Where does it run?

Symbolator 6 runs in the TI-89 calculator, and compatible devices. Symbolator does not run in the TI-Nspire machines, since they are incompatible with the TI-89 platform.

Why would I use it?

Symbolator is extremely useful in solving a wide variety of circuits theory problems, such as those taken by engineering students in Circuits I and Circuits II classes. Its main advantage is that Symbolator allows the student to focus in the conceptual understanding of circuit analysis, rather than the mathematical technics used for their solution.

Symbolator is a fantastic tool when you need a symbolic approach to a small or medium sized circuit composed of ideal, linear elements. You should use Symbolator whenever you have to manage symbolic values or need symbolic results. When compared to SPICE, PSpice or Electronic Workbench, this small software offers a simpler way to define dependent sources, and includes special elements such as ideal transformers and six different two-ports, usually not included in other simulators.

When is it most useful?

Symbolator is most useful with linear circuits with symbolic values, such as you find in undergraduate courses in circuit analysis. Students of Circuits I and II may benefit a lot from this pocket-sized circuit wizard for the numerical or symbolic solution of linear circuits. It can help you with learning circuits theory, doing your homework or taking that final exam. It is not, however, a replacement for your brain or an excuse to not study your circuit theory classes: you must understand circuits theory to use it.

Symbolator is a linear circuit simulator, and does not simulate non-linear elements, such as diodes and transistors.



How much does it cost?

Nothing. It's free. Has always been, will always be. (*You are welcome!*) Furthermore, since June 2013, Symbulator is offered under a Creative Commons BY-NC-SA license. If you want to port it to other platforms, you are free to do so under the CC license above. I look forward to ports to the TI-Nspire platform and to Mathematica!

Who made this thing?

I did. I started writing it in April 1999, as a junior in Electrical Engineering at Universidad Tecnológica de Panamá (UTP). Version 5 was done by January 2001. Symbulator won 1st place in the 2k IEEE Student Paper Contest - Latin America. It got me a Distinguished Alum award from UTP in 2008, and was at least part of the reason for the Outstanding Young Person award that the Panamanian Chapter of the Junior Chamber International gave me in 2010. In June 2013 I released a beta of version 6.

Get your calculator ready

*Get the current
versions of
Symbulator and
DiffEq online*

Symbulator relies on a software for Laplace transforms called DiffEq, made by Lars Frederiksen. You can download zip files for both Symbulator and DiffEq from this URL:

download.symbulator.com

Inside the zips you will find two files you need to transfer to your TI-89 calculator. For the transfer, following the manufacturer's instructions. Both Symbulator and DiffEq run faster when they are archived. To archive them, execute these two programs:

*Install the
programs so they
run faster*

`s\install()` and `dif\install()`

That is all! You are now all set. We'd love to hear from you. Please send us your message to **email@symbulator.com**. Have fun!

Running a simulation

*s\dc is the gate
to run a DC
simulation*

You will learn how to simulate through examples. For now, let me just say that we give Symbulator the circuit description as a string of elements separated by semi-colons.

The description can be stored in a variable or passed directly as an argument. To run a DC simulation in Symbulator, we use an access program, or "gate", called `s\dc`, which takes one argument: the circuit description in string form. Below you will find plenty of examples of DC simulations.

I recommend that you make MAIN your current folder and empty it before each simulation. Symbulator runs faster and more accurately if you use integers and fractions in your element values, instead of decimals. It is preferable to simulate with exact numbers and then evaluate the answers using `[≈]`, e.g. `♦[ENTER]`. In this book, when we say that we 'evaluate' an expression, what we mean is that we use either `[=]` or `[≈]` as required. Ok, let's get right into it!

Elements: *e and r*

Voltage source and resistors

Describing a resistor

In Symbulator, an ideal **resistor** is described as follows: *first* the name of the resistor, which must start with the letter **r**, coma; *second* the name of the first node of the resistor, another coma; *third* the name of the second node of the resistor, another coma; and *fourth* the value of the resistor (in Ω). There is no coma at the end.

For example, an ideal resistor called **r1**, connected between nodes **a** and **b** with a resistance of **300 Ω** , would be described thus: **r1,a,b,300** Since the value of resistors is often given in kilo ohms, Symbulator will interpret any **k** in the value of a resistor as an indication that the values are in k Ω , as long as the **s\si** variable is set to **true**. For example, a value of 8k Ω can be entered in many ways: **8000**, **8k** and **8E3** are equivalent. The only difference is that the first will be treated as exact values, while the others will be treated as approximate.

This is how resistors are described in Symbulator.

*When the **s\si** variable is true Symbulator interprets any **k** in the value of the resistor to mean kilo (e.g. x1000)*

Simulation answers

After the simulation in DC is complete, Symbulator stores a series of answers in the calculator's memory, labeled with easy to remember names for your convenience.

- The *voltage of each of its nodes* with reference to the ground is stored in a variable called **v** and the name of the node. For example, for a node called **1**, its voltage with reference to ground is calculated and stored in a variable called **v1**.
- The *voltage drop in the resistor*, that is to say the voltage in the first node minus the voltage in the second node, is stored in a variable called **v** and the name of the resistor. For example, for a resistor called **r5**, the voltage drop is stored in **vr5**.
- The *current through the resistor*, flowing from the first node towards the second, is stored in a variable called **i** and the name of the resistor. I.e., the current flowing through resistor **rx**, from the first node towards the second, is stored in **irx**.
- The *power consumed by the resistor* is stored in a variable called **p** and the name of the resistor. For example, the power consumed by resistor **r12** is stored in **pr12**.

What about conductances?

As an ideal circuit element, a conductance is not different from a resistor: they are the same element, with the only difference that their values are given in different terms. Resistance values are given in ohms and conductance values are given in siemens. A conductance value can be converted into an ohms value by dividing 1 over the conductance in siemens. Thus, for example, an 8S conductance between nodes a and b could be described thus as a 1/8 Ω resistor: **r1,a,b,1/8**

Describing an ideal voltage source

This is how voltage sources are described

Symbulator interprets any k or m in the value of a voltage source to mean kilo and milli, respectively, if s\si is true.

In Symbulator, an ideal **voltage source** is described as follows: *first* the name of the source, which must start with the letter **e**, a coma; *second* the name of the positive node of the source, another coma; *third* the name of the negative node of the source, another coma; and *fourth* the value of the source (in volts). No coma at the end.

For example, an ideal voltage source called **e1**, connected between nodes **3** and **0**, with a voltage of **12V** (given as voltage of node 3 with regards to node 0), would be described thus: **e1,3,0,12** Since the value of voltage sources is sometimes given in kV or mV, Symbulator will interpret any **k** or **m** in the value of a voltage source as an indication that the values are in kV or mV respectively, as long as **s\si** is set to **true**:

- For example, a value of 4kV can be entered as **4000**, **4k** and **4E3** are equivalent. The first is treated as an exact value.
- Another example: a value of 180mV can be entered as: **180m**, **180/1000**, **0.180**, **18E-2** and **180E-3**. The fraction will be treated as an exact value.

Simulation answers

After a DC simulation, for each voltage source in a circuit, Symbulator will store a series of answers in the current folder of the calculator:

- The *voltage of each of its nodes* with reference to the ground is stored in a variable called **v** and the name of the node. For example, for a node called **1**, its voltage with reference to ground is calculated and stored in a variable called **v1**.
- The *voltage drop in the source*, that is to say the voltage in the first node minus the voltage in the second node, is stored in a variable called **v** and the name of the source. For example, for a source called **e5**, the voltage drop is stored in **ve5**.
- The *current through the source*, flowing from the first node towards the second node, is stored in a variable called **i** and the name of the source. For example, the current flowing through a source called **ex**, from its first node towards its second node, is stored in **iox**. The way the current direction is defined might seem counterintuitive for voltage sources; it is such for consistency: the same direction is applied to all other two-node elements throughout Symbulator.
- The *power consumed by the source* is stored in a variable called **p** and the name of the source. For example, for a source called **e12**, the power consumed is stored in **pe12**; the power delivered is the negative of that, and can be found via **-pe12** (where - is the negative sign, not the subtraction operator.) For sources, the choice to store the consumed power instead of the delivered power may seem odd; it is such for consistency: for all elements, the power given is the consumed power.
- The *equivalent resistance* of the rest of the circuit, as seen by a source, is stored in a variable called **r** and the name of the source. For example, the equivalent resistance of a circuit as seen from a source called **e2** is stored in variable **re2**.

Illustrative examples

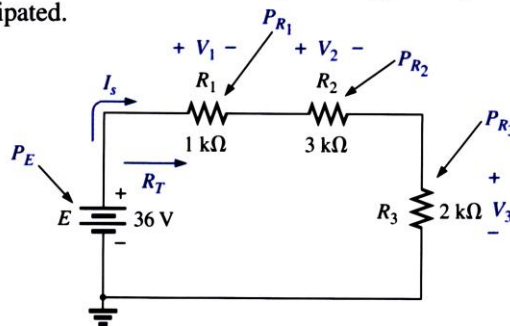
Solving a numerical circuit

Exemplum docet. Let's dive right in and solve a numerical linear circuit in direct current using the `s\dc` gate. A circuit is **numerical** if we know the numerical value of every element in the circuit. I have chosen Example 5.7 from Boylestad's *Introductory Circuit Analysis* (11e). Moving forward, I will refer to that textbook as **B11**. For your benefit, the problem statement and the circuit schematic were scanned and are reproduced below exactly as they appear in the textbook.

A numerical analysis is possible when we have the values of all the elements in the circuit.

B11's Example 5.7

- Determine the total resistance R_T .
- Calculate the current I_s .
- Determine the voltage across each resistor.
- Find the power supplied by the battery.
- Determine the power dissipated by each resistor.
- Comment on whether the total power supplied equals the total power dissipated.



Step 1: Describe the circuit

The first step is to describe the circuit. Circuit description starts with naming the nodes. You can call the nodes anything you want, be it a number or a letter, as long as the name is unique. You must always have one node called **0** (zero); this is your ground node and has a voltage of 0V. I labeled the nodes starting in the ground and moving clockwise: **0**, **1**, **2** and **3**. It helps me to pencil the names in the schematic itself.

After naming the nodes, I am ready to describe the circuit in Symulator notation.

When I only have one voltage source, like here, I enjoy naming it with a single letter: **e**. I describe the voltage source as follows: `e,1,0,36` since its name is **e**, its positive node is called **1**, its negative node is called **0**, and its value is 36 volts between these nodes.

Now I describe the **resistors**. The first resistor I described as follows: `r1,1,2,1k` because I name it **r1**, its first node is called **1**, its second node is called **2**, and its value is 1k ohms. The second resistor we describe similarly: `r2,2,3,3k` and likewise for the third resistor: `r3,3,0,2k` Since we want to use k, set the SI flag to true, by typing:

`true→s\si`

First step is to describe the circuit. Start by naming the nodes.

Your ground should always be node 0



The circuit is passed to Symbulator as a string, elements separated by colons

We pass along this description to Symbulator as a string, e.g. we open with a quotation mark, then enter the descriptions of each element, separated with colons, and then close with a quotation mark. We can store this string in a variable. I like to call this variable `cir` (short for circuit,) but you can call it something else if you prefer. I type this in my Symbulator-ready calculator:

```
"e,1,0,36:r1,1,2,1k:r2,2,3,3k:r3,3,0,2k"→cir
```

Once I press enter, the circuit will be stored in that variable.

Step 2: Run the simulation

We now ask Symbulator to simulate this circuit in direct current (DC), by typing this:

```
s\dc(cir)
```

After you press enter, Symbulator will quickly simulate the circuit stored in `cir`. Symbulator will let you know when it is *Done*. It took 12 seconds in my calculator to get 16 answers, and store them in variables in the current folder: the voltage in each of the three nodes (`v1`, `v2` and `v3`), for each of the four elements in the circuit its voltage drop (`ve`, `vr1`, `vr2` and `vr3`), current (`ie`, `ir1`, `ir2` and `ir3`) and power consumed (`pe`, `pr1`, `pr2` and `pr3`), and the equivalent resistance as seen by the source (`re`).

Some alternatives

There is more than one way to skin a cat...

We could also have given Symbulator the command to simulate and the circuit description in a single line, using two possible methods. The first is connecting the two steps above using a colon, as shown below:

```
"e,1,0,36:r1,1,2,1k:r2,2,3,3k:r3,3,0,2k"→cir:s\dc(cir):
```

The second is giving the circuit description as an argument of the `s\dc` gate:

```
s\dc("e,1,0,36:r1,1,2,1k:r2,2,3,3k:r3,3,0,2k")
```

Any of the three approaches yields the same result, and I use them interchangeably.

Step 3: Get the answers

Answer to question (a). The equivalent resistance seen by the source `e` is stored in: `re`. Evaluate this expression. The calculator returns the right answer: **6000** (e.g. 6KΩ.)

Answer to question (b). Current I_s is the current flowing through the source from node 0 to node 1; thus, we can obtain it by evaluating `-ie`. Since this is a series circuit, we could also find it as the current through any of the resistors: `ir1`, `ir2` or `ir3`. In either case, the calculator provides the right answer: **.006** (e.g. 6mA.)

Answer to question (c). The voltage drop in resistor R_1 is found evaluating `vr1`: the calculator returns **6** (e.g. 6 V.) For R_2 , evaluate `vr2` to obtain **18** V. And for R_3 , `vr3` gives us **12** V. These are all the right answers.

Answer to question (d). We evaluate `-pe` and get the right answer: **.216** (e.g. 216mW)

Answer to question (e). The power consumed by the resistors are found evaluating `pr1`, `pr2`, and `pr3`. The calculator returns the right answers: **.036** (e.g. 36mW), **.108** (e.g. 108mW), and **.072** (e.g. 72mW) respectively.

Answer to question (f). Let's ask the calculator if the sum of the consumed power in the resistors equals the power supplied by the source. Evaluate this equation:

`pr1+pr2+pr3=-pe`

The calculator returns **true**. This is the right answer, and concludes the solution.

An alternative

You can ask for all the answers above in a single line, by using an array, such as this:

`{re,ir1,vr1,vr2,vr3,-pe,pr1,pr2,pr3,pr1+pr2+pr3=-pe}`

We evaluate this array, and get the same answers as before, also as an array. Actually, we could solve the whole problem using a single line of text, as follows:

`true→s\si:s\dc("e,1,0,36:r1,1,2,1k:r2,2,3,3k:r3,3,0,2k"):
approx({re,ir1,vr1,vr2,vr3,-pe,pr1,pr2,pr3, pr1+pr2+pr3=-pe})`

An easy way to read this is to identify the colons as delimiters between things that you would have asked in separate lines. First, set the SI flag to true; another colon; second, the command to simulate in DC the circuit provided between quotation marks; another colon; and finally an array of all the answers we would like to evaluate.

By showing you this, I just want to make you aware of the possibilities. But please do not rush into trying to solve a whole problem using a single line. There are times when you may not be sure about your circuit description or about which variables to evaluate as answers. In these cases, it is easier to solve the problem step by step: first describe the circuit, then run the simulation and finally find the answers.

Practice problems of this type are found starting in page 112.

A word on symbolic problems

I call a circuit **symbolic** if one or more of the element has an unknown value. It is the ability to simulate symbolic circuits that gives Symbulator its name and sets it apart from other programs. I distinguish between two types of symbolic simulations.

Purely symbolic problems

The first type is when the desired answers are also symbolic, e.g. given as algebraic functions in terms of unknown values (what I call a *purely symbolic* problem.)

Numerical-from-symbolic problems

The second type is when numerical answers are expected from a symbolic circuit (what I call a *numerical-from-symbolic* problem.) This solution is possible when they

*We can evaluate
multiple
answers using
an array*



Symbulator has an Expert mode for faster and better results in symbolic to numeric problems

give us additional information about the circuit. Numerical-from-symbolic problems can be solved in two ways in Symbulator:

- If they are simple, e.g. if there's just one or two unknown values in the circuit, and we are only asked for one or two numerical answers, we can simulate the circuit symbolically and then solve for the numerical answers using `solve()`
- If they are not simple, e.g. if there are many unknown values in the circuit, or we are asked for many numerical answers, it is easier and more time-efficient to use the Expert mode of Symbulator, whose gate is `slex(cir)`

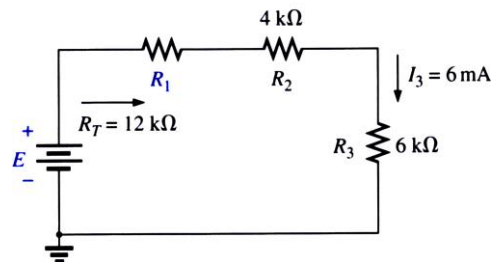
Let's see an example of each approach to solving numerical-from-symbolic problems.

Solving a symbolic circuit with *solve*

Let's do a numerical-from-symbolic problem by means of the `solve` command. We will obtain numerical values from the simulation of a circuit with some unknown element values. Later we will solve this same problem using the expert mode.

B11's Example 5.6

Given R_T and I_3 , calculate R_1 and E



This is a very nice symbolic problem: we should be able to solve it into numerical results because, even though the problem hides two values from us (e.g. the value of the source E and of the resistor R_1), it gives us in exchange two answers (e.g. the equivalent resistance R_T and the current I_3) that we can use to solve for the unknowns.

Unknown values can be entered as a variable.

Just make sure the variable is empty!

Since this circuit is structurally identical to B11's Example 5.7, we will use the same names for the nodes. The circuit description is identical except for the element's values. As values for the elements in the circuit, Symbulator will accept numbers, variables or even algebraic expressions. For this example, I chose to use `e` for the value of source `e` and `r1` for the value of the `r1` resistor¹. Make sure that the variables are empty, either by emptying the current folder or by deleting them from the calculator's memory, thus: `DelVar e,r1`. Set the SI flag: `true→s\si`. Below is how I described this circuit. I pass along this description to Symbulator as a string, and store it in a variable.

`"e,1,0,e:r1,1,2,r1:r2,2,3,4k:r3,3,0,6k"→cir`

¹ Giving a symbolic element a value equal to its name is a useful trick to remember whose value it is. However, due to the potential for conflicts, reserved names should be avoided in the expert mode, which stores all solved unknowns into variables, as the third example shows.

Ask Symbulator to simulate this circuit in direct current (DC), by typing this:

```
s\dc(cir)
```

A moment later (16 seconds in my calculator²), Symbulator is *Done*, and we are ready to answer the questions. Using the symbolic answers provided by Symbulator and the known answers given by the problem, we can write two equations. We can then solve these two equations for the two unknowns that interest us, using the calculator's **solve** command.³ Let's explore what we have here.

The problem says that I_3 is 6mA. In Symbulator, I_3 is called **ir3** (e.g. the current through resistor **r3**). If you evaluate **ir3**, you will see it produces an algebraic expression in terms of the two unknowns, **e** and **r1**. This is what we call a symbolic answer. The problem also says that R_T is 12K Ω . As we saw in the previous problem, the equivalent resistance as seen by the source **e** is given by **re**, which when evaluated gives another algebraic expression in terms of **r1**. We can then write two new equations, **re=12000** and **ir3=6/1000**, and solve them for the two unknowns we want: **e** and **r1**.

```
solve(re=12000 and ir3=.006,{e,r1})
```

An instant later we get the answers: **e = 72** (volts) and **r1 = 2000** (ohms). These are the right answers. Not many other circuit simulators allow this flexibility.

Examples to practice this type of problems are found starting in page 116.

Notice the kilo nomenclature does not apply outside of the circuit description

Solving a symbolic circuit with *Expert*

Symbulator's true strength is seen in numerical-from-symbolic problems like the one we solved above, but using its Expert mode of simulation, which cracks these problems open even faster and can give you fully numerical answers to problems like this with an equal number of unknown values and of answers provided by the problem. Learning to use the expert mode pays off handsomely in terms of additional power and speed. Let's solve the same circuit from **B11**'s Example 5.6, this time using the expert mode.

B11's Example 5.6 – *Redux*

We will use the same circuit description as before, with a single change: we will use **rx** for the value of resistor R_1 , instead of the **r1** value we used before. Like this:

```
"e,1,0,e:r1,1,2,rx:r2,2,3,4k:r3,3,0,6k"→cir
```

The reason for this is that **r1** is a reserved variable in the calculator. The expert mode tries to store all the values of the solved unknowns into variables of the same name. Trying to store a value into **r1** would result in an error.

² If you want to know how long it took Symbulator to run a simulation, you have to make sure your calculator's clock is on, by typing **ClockOn**. As long as the clock is on, you can see how long a simulation took by going to the Program IO screen (press F5 in the main menu.)

³ If you are not familiar with the solve command, I refer you to the calculator's manual.



Set the SI flag and run the expert mode by typing this in the calculator:

true→s\si

s\ex(cir)

*Variables **r1** through **r99** are reserved variable in the calculator. This means you cannot store things into them.*

When prompted, select **DC** and press Enter. Now you will see a prompt asking you to add equations, variables and conditions. You may recall from your algebra class that you need an equal number of equations and unknowns in order to solve a set of equations into numerical values. The statement of the problem gives us the information we need to write the two additional equations⁴. In "Add equations" type:

re=12k and ir3=6m

If you want to get numerical answers, you must have as many equations as you have unknowns

In "Add unknowns" type:

e,rx

Now we have six variables and six equations. Press Enter and wait just a few seconds. A few of other dialogs will appear. In this and all the other Expert examples in this volume, just press OK in these prompts without changing anything in them.

When Symbolator says "Done", go ahead and retrieve the answers:

rx

You get **2000**, the right answer.

e

You get **72**, the right answer.

The speed advantage of the expert mode is not necessarily evident in this simple problem. It does give you an idea of what the expert mode is all about: you get to halt the simulation in mid-air and give Symbolator extra information. Had this circuit been larger, the benefit of the expert mode in computation time would be clear.

Examples to practice this type of problems are found starting in page 121.

Practice problems

Numerical problems

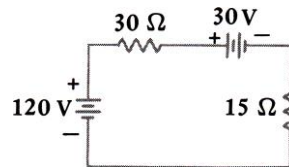
The following are practice problems taken from several textbooks. They were chosen because they apply only the concepts that you have learned so far. This will allow you to practice these concepts and reinforce them before moving on to new ones.

The problem below comes from Figure 1-26 (a) in Hyat and Kemmerly's *Engineering Circuit Analysis* (5ed). Moving forward, we will refer to that textbook as **HK5**.

⁴ We know the value for R_T and the value for I_S in terms of the variables listed in the prompt: because R_T is given by r_e , which is a function of **e** and **ie**, and I_S is given by ir_3 , which is a function of **ie**.

HK5's Figure 1-26

We are asked to determine the current, voltage drop and power consumed in each resistor, as well as the power *delivered* by each voltage source. We are also asked to check that the powers in the circuit add up to zero. Here is my solution.



I named the nodes thus: the bottom node is named **0**, the top nodes, from left to right, are named **1**, **2** and **3**. My description of the circuit is given below, followed after a colon by the DC simulation command.

```
s\dc("e1,1,0,120:r1,1,2,30:e2,2,3,30:r2,3,0,15")
```

When the simulation is done, you can ask the calculator for the answers you need:

- Evaluating **ir1** or **ir2** gets the current in the resistors: **2 A**
- Evaluating **vr1** gets the voltage drop in the 30Ω resistor: **60 V**
- Evaluating **vr2** gets the voltage drop in the 15Ω resistor: **30 V**
- Evaluating **pr1** gets the power consumed in the 30Ω resistor: **120 W**
- Evaluating **pr2** gets the power consumed in the 15Ω resistor: **60 W**
- Evaluating **-pe1** gets the power delivered by the 120V source: **240 W**
- Evaluating **-pe2** gets the power delivered by the 30V source: **-60 W**. This means this source is actually consuming 60W.
- Evaluating **pr1+pr2+pe1+pe2** gets the sum of powers: **0 W**. As expected.

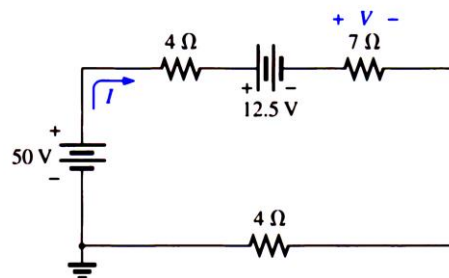
Wasn't that easy? We could also have asked for all the answers with one array:

```
{ir1,vr1,vr2,pr1,pr2,-pe1,-pe2,pr1+pr2+pe1+pe2}
```

B11's Example 5.20

The practice problems will get progressively more complicated as we move on. This will allow you to build up your '*symbolating*' skills with confidence.

Determine I and the voltage across the 7 Ω resistor



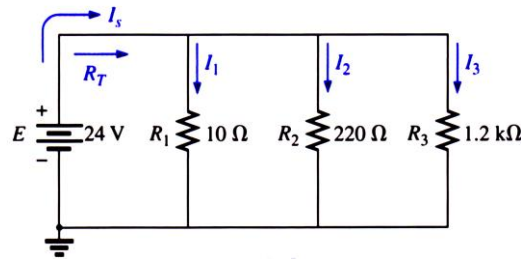
I named the nodes clockwise starting from the ground: **0**, **1**, **2**, **3** and **4**. Below is my circuit description, given as an argument of the command to do the DC simulation:

`s\dc("e1,1,0,50:r1,1,2,4:e2,2,3,12.5:r2,3,4,7:r3,4,0,4")`

When it's done, ask for the answers we need. Evaluating `ir1` gets the current I : **2.5 A**.
Evaluating `vr2` gets the voltage drop in the 7Ω resistor: **17.5 W**

B11's Example 6.13

- Find the total resistance.
- Calculate the source current.
- Determine the current through each branch.



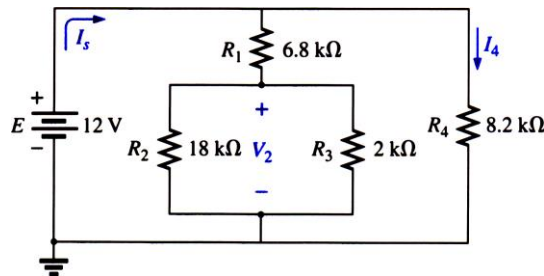
Since the ground is the bottom node, I named it **0**. I named the top node **1**.

`s\dc("e,1,0,24:r1,1,0,10:r2,1,0,220:r3,1,0,1.2k")`

- Evaluating `re` gets the total resistance: **9.49 Ω**
- Evaluating `-ie` gets us the source current: **2.53 A**
- Evaluating `ir1` gets I_1 : **2.4 A**, `ir2` gets I_2 : **0.11 A**, and `ir3` gets I_3 : **0.02 A**.

B11's Example 7.2

Determine I_4 , I_s and V_2 . My solution below. I named the top node **1**, and the other **2**:

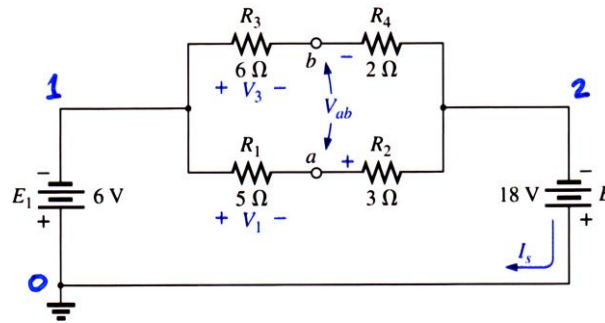


`true→s\si:s\dc("e,1,0,12:r1,1,2,6.8k:r2,2,0,18k:r3,2,0,2k:r4,1,0,8.2")`

Answers: V_2 is **2.51 V**, `-ie` (e.g. I_s) is **2.86 mA** and `ir4` is **1.46 mA**.

B11's Example 7.7

- Find the voltages V_1 , V_3 , and V_{ab} for the network
- Calculate the source current I_s .



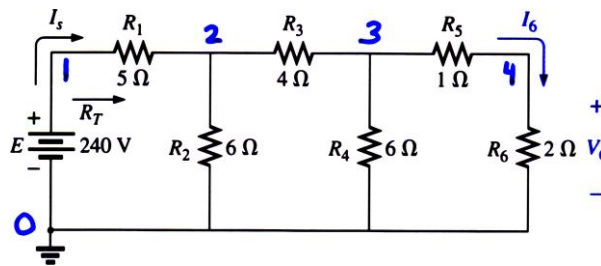
For your benefit, I have labeled the node names I used. My solution:

```
s\dc("e1,0,1,6:e2,0,2,18:r1,1,a,5:r2,a,2,3:r3,1,b,6:r4,b,2,2")
```

Answers: V_1 is **7.5 V**, V_3 is **9 V**. For V_{ba} , $V_b - V_a$ is **-1.5 V**. For I_s , $-i_{e2}$ is **3 A**.

B11's Figure 7.32

Determine I_6 and V_6 . My solution below:

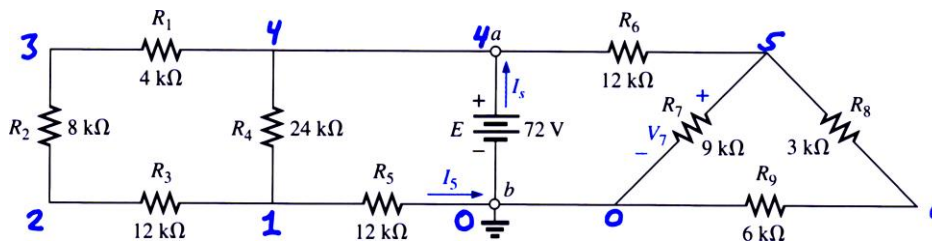


```
s\dc("e,1,0,240:r1,1,2,5:r2,2,0,6:r3,2,3,4:r4,3,0,6:r5,3,4,1:r6,4,0,2")
```

Answers: I_6 is **10 A**, and V_6 is **20 V**.

B11's Example 7.10

Calculate the indicated currents and voltages.



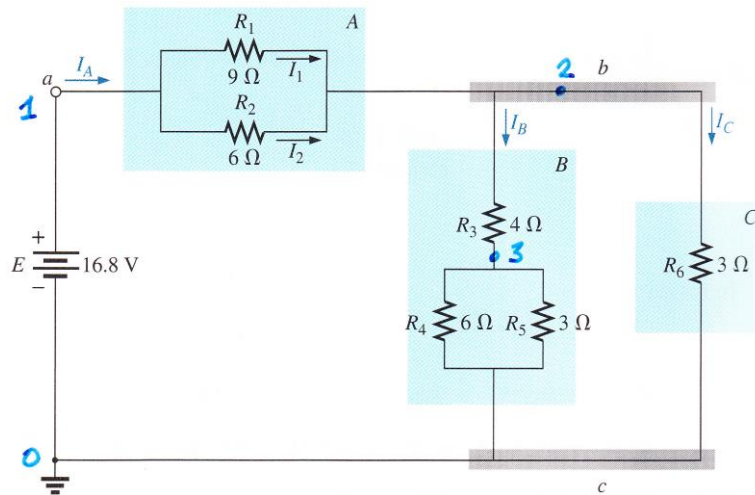
My solution below:

```
true→s\si:s\dc("r2,2,3,8k:r1,3,4,4k:r3,1,2,12k:r4,1,4,24k:r5,1,0,12k:e,4,0,72:r6,4,5,12k:r7,5,0,9k:r8,5,6,3k:r9,0,6,6k")
```

Answers: I_5 is **3 mA**, $-i_{e5}$ (e.g. I_5) is **7.36 mA**, and V_7 is **19.6 V**.

B11's Example 7.4

Determine the currents I_1 , I_2 , I_A , I_B and I_C , and the voltage drop areas A, B and C.



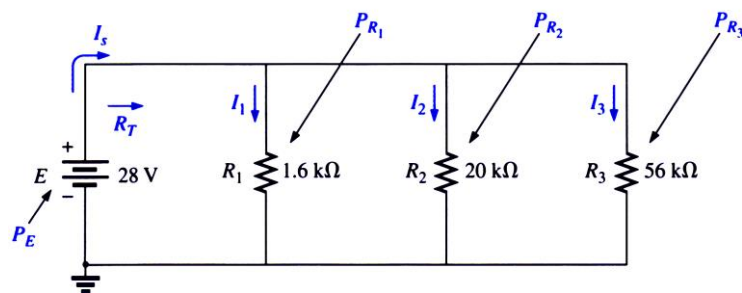
My solution below:

```
s\dc("e,1,0,16.8:r1,1,2,9:r2,1,2,6:r3,2,3,4:r4,3,0,6:r5,3,0,3:r6,2,0,3")
```

Current I_1 is found via $ir1 = 1.2$ A, I_2 via $ir2 = 1.8$ A, I_A via $-ie = 3$ A, I_B via $ir3 = 1$ A and I_C via $ir6 = 2$ A. The voltage drop in area A is $vr1 = 10.8$ V; in both B and C it is $v2 = 6$ V.

B11's Example 6.15

- Determine the total resistance R_T .
- Find the source current and the current through each resistor.
- Calculate the power delivered by the source.



My solution below:

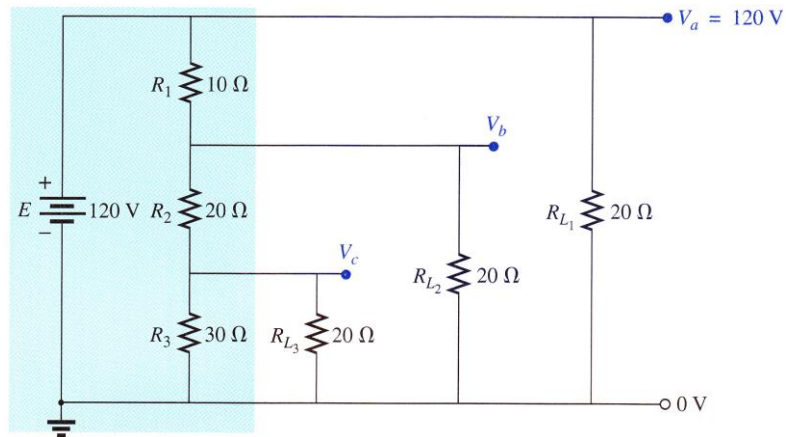
```
s\dc("e,1,0,28:r1,1,0,1.6k:r2,1,0,20k:r3,1,0,56k")
```

- Evaluating re gets the total resistance: **1.44** kΩ
- Evaluating $ir1$ gets **17.5** mA, $ir2$ gets **1.4** mA, and $ir3$ gets **0.5** mA
- Evaluating $-pe1$ gets the power: **543** mW

These are the correct answers.

B11's Figure 7.40

Determine V_b and V_c .



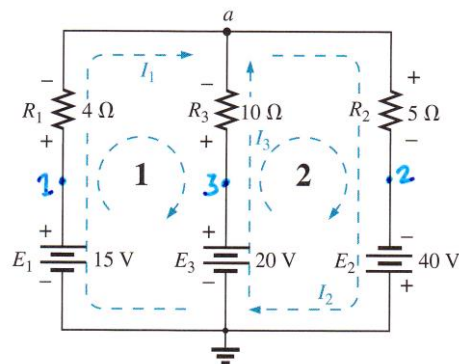
My solution below:

```
s\dc("e,a,0,120:r1,a,b,10:r2,b,c,20:r3,c,0,30:
r11,a,0,20:r12,b,0,20:r13,c,0,20")
```

Answers: v_b is **66.21 V**, and v_c is **24.83 V**.

B11's Example 8.10

Determine the current through each resistor.



My solution below:

```
s\dc("e1,1,0,15:r1,1,a,4:e3,3,0,20:r3,3,a,10:e2,0,2,40:r2,a,2,5")
```

Through an array, and using the `approx` command, we ask for all the three answers:

```
approx({ir1,ir2,ir3})
```

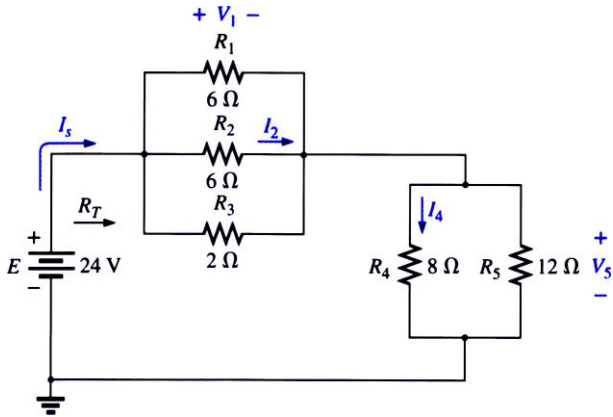
The calculator returns **{4.77,7.18,2.41}** meaning $I_{R1} = 4.77A$, $I_{R2} = 7.18A$ and $I_{R3} = 2.41A$.

These are the correct answers. We could get this answer in a single-line command:

```
s\dc("e1,1,0,15:r1,1,a,4:e3,3,0,20:r3,3,a,10:e2,0,2,40:r2,a,2,5"):approx({ir1,ir2,ir3})
```

Moving forward, we will often use this single-line approach for getting our answers.

B11's Example 7.6



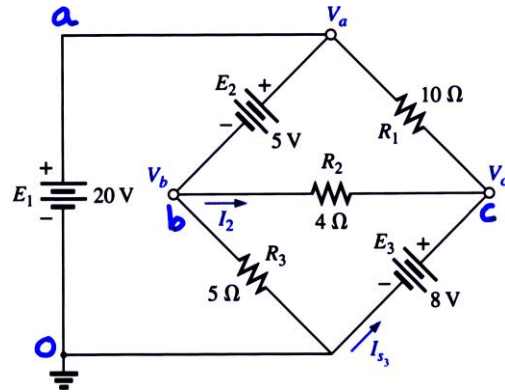
My solution below:

`s\dc("e,1,0,24:r1,1,2,6:r2,1,2,6:r3,1,2,2:r4,2,0,8:r5,2,0,12")`

Answer: -ie is $I_s=4$ A, ir2 is $I_2=.8$ A, ir4 is $I_4=2.4$ A, vr1 is $V_1=4.8$ V, vr5 is $V_5=19.2$ V.

B11's Example 7.11

- Determine voltages V_a , V_b , and V_c .
- Find voltages V_{ac} and V_{bc} .
- Find current I_2 .
- Find the source current I_{s3} .



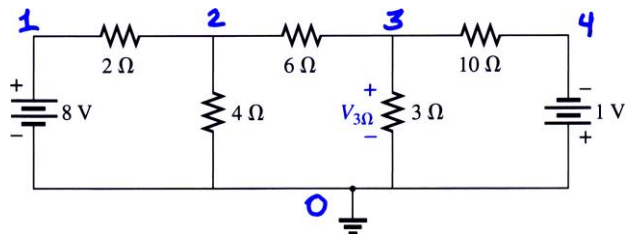
My solution below:

`s\dc("e1,a,0,20:e2,a,b,5:e3,c,0,8:r1,a,c,10:r2,b,c,4:r3,b,0,5")`

Answers: $v_a=20$, $v_b=15$, $v_c=8$, $v_a-v_c=12$, $v_b-v_c=7$, $i_{r2}=1.75$, I_s via -ie3=-2.95

B11's Example 8.24

Find the voltage drop in the 3Ω resistor.



My solution below:

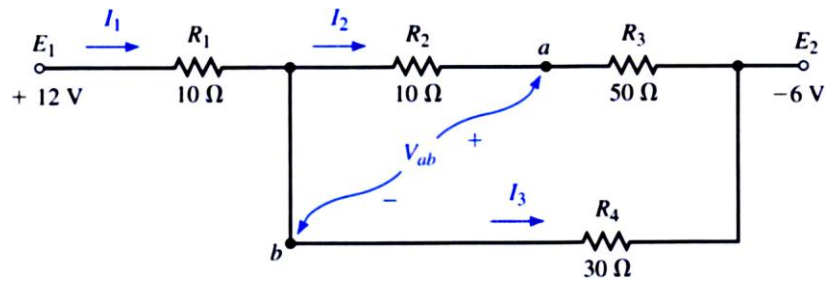
`s\dc("e8,1,0,8:r2,1,2,2:r4,2,0,4:r6,2,3,6:r3,3,0,3:r10,3,4,10:e1,0,4,1")`



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RM3's Example 7-5 (Hidden source)

Find the indicated currents and voltages.



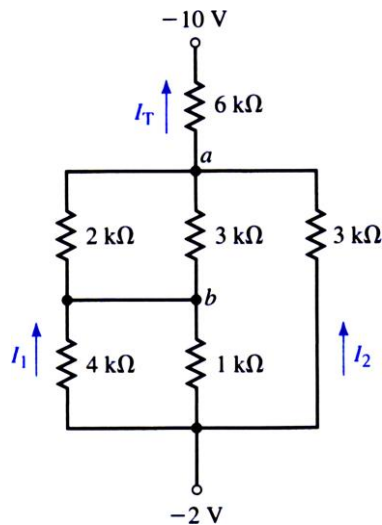
In my solution below, notice I introduced two sources, one for 12V and one for -6V:

```
s\dc("e1,1,0,12:r1,1,b,10:r2,b,a,10:r3,a,2,50:r4,b,2,30:e2,2,0,-6"):approx({ir1,ir2,ir4,va-vb})
```

The answer, **{.6,.2,.4,-2.}**, indicates $I_1=.6$, $I_2=.2$, $I_3=.4$ and $V_{ab}=-2$. This is correct.

RM3's Figure 7-16 (Hidden source)

Find the total circuit resistance, and the indicated currents and voltages.



My solution below:

This is how I named the nodes: the top, **1**; the bottom, **2**, and **a** and **b** as in the figure. In my solution below, notice I introduced two sources, one for -10V and one for -6V:

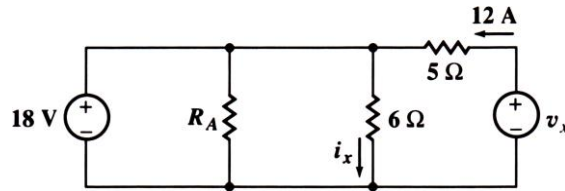
```
true→s\si:s\dc("r1,1,b,4k:r2,1,a,3k:r3,b,a,2k:r4,b,a,3k:r5,1,b,1k:rt,a,2,6k:e1,1,0,-2:e2,2,0,-10"):approx(({v1-v2)/irt,irt,ir1,ir2,va-vb})
```

The answer we get indicates that the equivalent resistance, given by $(v_1-v_2)/I_T$, is **7.2** KΩ, and that $I_T=$ **1.11 mA**, $I_1=$ **.133 mA**, $I_2=$ **.444 mA** and $V_{ab}=$ **-0.8 V**. This is correct.

Symbolic problems with *solve*

HK5's Figure 1-24a (*solve*)

Determine i_x and v_x in the following circuit.



I named the nodes thus: bottom is **0**, top left is **1**, top right is **2**. I named the elements according to their value: this facilitates remembering who's who in the circuit. I also defined the resistors' nodes in the direction of the current indicated in the diagram.

```
s\dc("e18,1,0,18:ra,1,0,ra:r6,1,0,6:r5,2,1,5:evx,2,0,vx")
```

Explore the answers. Since **ir5** (which we know is 12A) is in terms of v_x , we can find v_x :

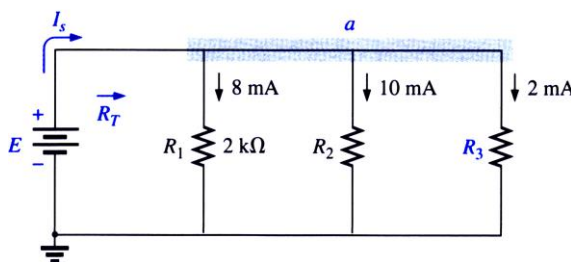
```
solve(ir5=12,vx)
```

We get that v_x is **78 V**, which is correct. Evaluating **ir6** we find that i_x is **3 A**. The fact that we can find numerical answers in this problem can be quite puzzling until one realizes that ignoring the value of R_A doesn't matter: due to the circuit's structure, it is not needed it to answer the two questions we have been asked.

Symbolic problems with *Expert*

B11's Example 6.19 (*Expert*)

- Determine the source current I_s .
- Find the source voltage E .



This problem, having three unknown element values and three known answers, is a perfect candidate for the expert mode. Below is my circuit description.⁵

```
true→s\si:s\ex("e1,1,0,e:r1,1,0,2k:r2,1,0,rr2:r3,1,0,rr3")
```

Select DC. Add these three equations:

⁵ In the expert mode, we avoid using as values to be solved any variable between $r1$ and $r99$, which are *reserved variables*. We also avoid using an element's name as its symbolic value.

$i_{r1}=8\text{m}$ and $i_{r2}=10\text{m}$ and $i_{r3}=2\text{m}$

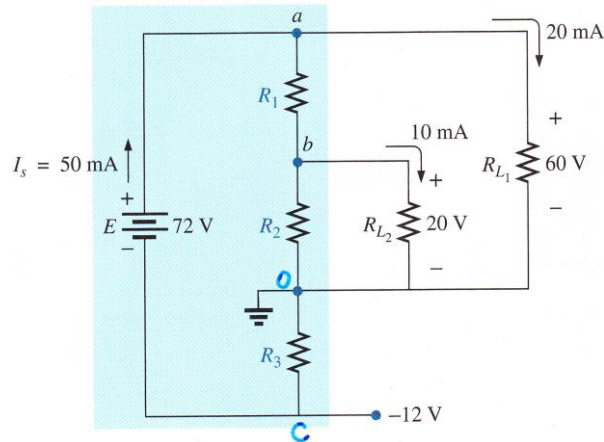
Add these unknowns:

e, r_2, r_3 :

Press Enter and Enter. When Symbolator is *Done*, find I_s and E by asking: `approx({-ie1,e})` We get that I_s is **0.2** A and E is **16** V.

B11's Example 7.12 (Expert)

Determine R_1 , R_2 and R_3 for the voltage divider supply. Can 2W resistors be used?



This problem is also perfect for the expert mode, because: (a) the target is to obtain numerical values, and (b) we have N unknown element values, and in turn we are given N numerical answers. Here is how I solved it:

```
true→s\si:s\ex("e,a,c,72:r1,a,b,rr1:r2,b,0,rr2:
r3,0,c,rr3:rl1,a,0,rrl1:rl2,b,0,rrl2")
```

Choose DC. Add these five equations:

$i_{rl1}=20\text{m}$ and $v_{rl1}=60$ and $i_{rl2}=10\text{m}$ and $v_{rl2}=20$ and $-ie=50\text{m}$

Add these five variables to the list of first level variables:

$r_1, r_2, r_3, r_{l1}, r_{l2}$

Enter. When Done, evaluate these variables to find the answers: $r_1 = 1.33\text{ K}\Omega$, $r_2 = 1\text{ K}\Omega$, $r_3 = 240\text{ }\Omega$. These are correct. Since all powers consumed in the resistors p_{r1} , p_{r2} and p_{r3} are smaller than 2W, it is possible to use 2W resistors in the design.

Element: j

Current source

Describing an ideal current source

In Symbulator, an ideal **current source** is described as follows: first the name of the source, which must start with the letter **j**, and a coma, second the name of the first node, another coma, third the name of the second node, another coma, and fourth the value of the source (in amperes). No coma at the end. *The value should be given in terms of the current flowing through the source from the first node towards the second node.* This means that the value of the source is how much current leaves the source out of the second node, and also how much current enters the source's first node.

An ideal current source called **j1**, connected between nodes **3** and **0**, with a current of **5A** (through it from node 3 towards node 0), would be described thus: **j1,3,0,5**. Since the value of current sources is often given in milli amps, Symbulator will interpret any **m** in the value of a current source as an indication that the values are in mA, as long as **s\si** is **true**. For example, a value of 2mA can be entered in many ways: **2/1000**, **8m** and **2E-3**. The only difference is that the first will be treated as an exact value.

Simulation answers

For each current source in a circuit, Symbulator will store a series of answers in the current folder of the calculator:

- The *voltage of each of its nodes* with reference to the ground is stored in a variable called **v** and the name of the node.
- The *voltage drop in the source*, that is to say the voltage in the first node minus the voltage in the second node, is stored in a variable called **v** and the name of the source. For example, for a source called **j5**, the voltage drop is stored in **vj5**.
- The *current through the source*, flowing from the first node towards the second node, is stored in a variable called **i** and the name of the source. For example, the current flowing through a source called **jx**, from its first node towards its second node, is stored in **ijx**. This may seem redundant, but it is actually rather handy.
- The *power consumed by the source* is stored in a variable called **p** and the name of the source. For example, for a source called **j12**, the power consumed is stored in **pj12**; the power delivered is the negative of that, and can be found via **-pj12**.
- The *equivalent resistance* of the rest of the circuit, as seen by a source, is stored in a variable called **r** and the name of the source. For example, the equivalent resistance of a circuit as seen from a source called **j2** is stored in variable **rj2**.

This is how current sources are described in Symbulator.

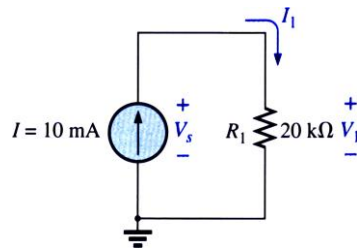
When s\si is true, Symbulator interprets any m in the value of the current source to mean milli (e.g.

Illustrative examples

A simple example: J and R

B11's Example 8.1

Given the circuit below, determine the current and voltage drop in R_1 .



```
true→s\si:s\dc("j,0,1,10m:r1,1,0,20k"): {ir1,vr1}
```

In the line above we are doing three things: the first is setting the SI flag to true, the second is asking Symulator to run a DC simulation of the circuit described between quotations, and the third is asking for the values of two variables: **ir1** and **vr1**.

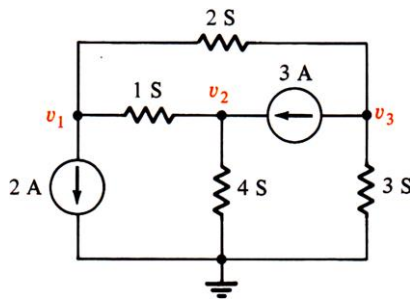
The calculator returns **{.01,200.}**, meaning a 10mA current and a 200V voltage drop.

An example with conductances

The following example is taken from the textbook *Elementary Linear Circuit Analysis* (2ed) by Leonard S. Bobrow. From this point forward, I will refer to this book as **Bo2**.

Bo2's Example 2.2

Given the circuit below, determine the voltages in the nodes.



As I explained before, in Symulator all conductances are simulated as resistors. So, the 4 siemens conductance becomes a 1/4 resistor, and so on. Below my description:

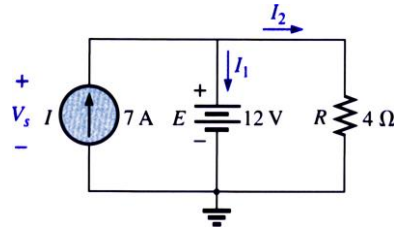
```
s\dc("j10,1,0,2:r12,1,2,1:r20,2,0,1/4:r30,3,0,1/3:
r13,1,3,1/2:j32,3,2,3"):approx({v1,v2,v3})
```

The answer, **{-1.3,.34,-1.12}**, indicates $v_1 = -1.3\text{V}$, $v_2 = .34\text{V}$, $v_3 = -1.12\text{V}$. This is correct.

An example with E, J and R

B11's Example 8.2

Determine the values of V_s , I_1 and I_2 .



In the line below, we concatenate three commands using colons. The first stores the circuit's definition in a variable. The second asks Symbolator to run a DC simulation of the circuit described in that variable. The third asks the calculator to provide us the values of three variables that – given the circuit description – answer the questions.

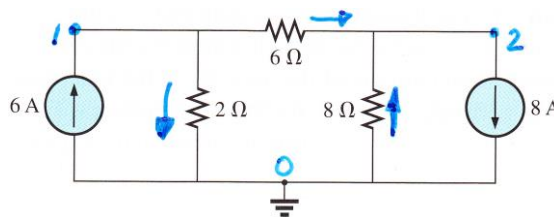
```
s\dc("j,0,1,7:e,1,0,12:r,1,0,4"): {v1,ie,ir}
```

The calculator returns **{12,4,3}**, meaning V_s is 12V, I_1 is 4A and I_2 is 3A. These are the correct answers. We will continue to use the single line instruction as we move on.

Practice problems

B11's Example 8.15

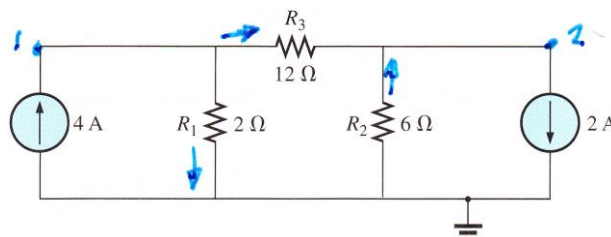
Determine the current through each resistor. My solution below, direction in blue:



```
s\dc("j6,0,1,6:r2,1,0,2:r6,1,2,6:r8,0,2,8:j8,2,0,8"):approx({ir2,ir6,ir8})
```

We get these answers: **{1.25,4.75,3.25}**. So I_{R2} is 1.25A, I_{R6} is 4.75A, and I_{R8} is 3.25A.

B11's Example 8.21



Determine the voltage in each node and the current through each resistor.

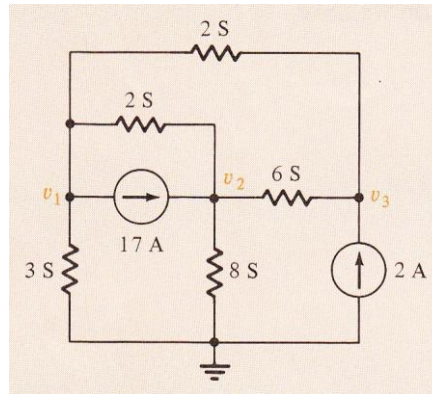
My solution below:

```
s\dc("j1,0,1,4:r1,1,0,2:r3,1,2,12:r2,0,2,6:
      j2,2,0,2"): {v1,v2,ir1,ir2,ir3}
```

We get the following answers: **{6,-6,3,1,1}**. So $V_1=6V$, $V_2=-6V$, $I_{R1}=3A$, and $I_{R2}=I_{R3}=1A$.

Bo2's Drill Exercise 2.2 (Conductances)

Given the circuit below, determine the voltages in the nodes.



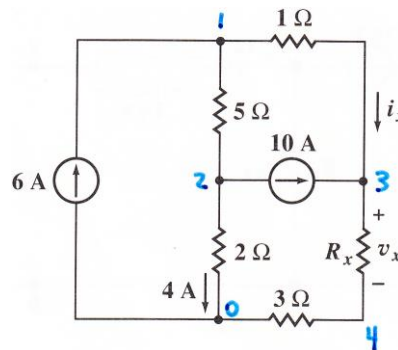
Below my solution:

```
s\dc("r10,1,0,1/3:r12,1,2,1/2:r13,1,3,1/2:r23,2,3,1/6:
      r20,2,0,1/8:j12,1,2,17:j03,0,3,2"):approx({v1,v2,v3})
```

The answer, **{-2,1,0.5}**, indicates $v_1=-2V$, $v_2=1V$, $v_3=0.5V$. This is correct.

HK5's Figure 1-24b (Expert)

Determine i_x and v_x in the following circuit.



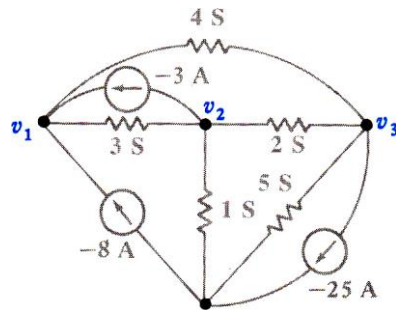
My solution below:

```
s\ex("j6,0,1,6:r5,1,2,5:r2,2,0,2:r1,1,3,1:
      r3,0,4,3:j10,2,3,10:r,3,4,rx"): {ir1,vr}
```

Select DC. Add equation $ir2=4$. Add unknown rx . Run the simulation. The answer, **{-8,80}**, means that I_x is -8A and that V_x is 80V.

HK5's Example 2.2 (Conductances)

Determine the voltages in the nodes. My solution below:

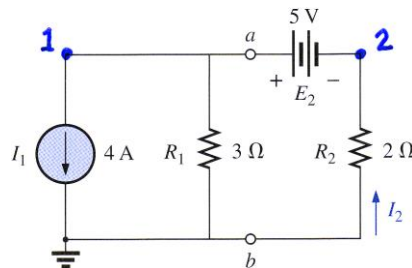


```
s\dc("j01,0,1,-8:j30,3,0,-25:j21,2,1,-3:r12,1,2,1/3:r23,2,3,1/2:r13,1,3,1/4:r20,2,0,1:r30,3,0,1/5"): {v1,v2,v3}
```

The answer, **{1,2,3}**, is correct: $v_1=1\text{V}$, $v_2=2\text{V}$, $v_3=3\text{V}$.

B11's Example 8.5

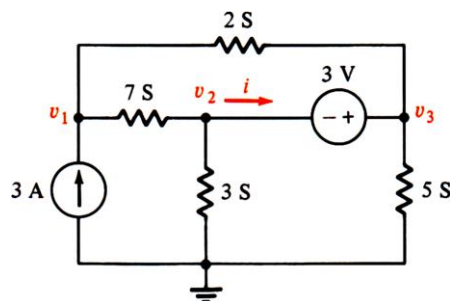
Determine the current I_2 . My solution below:



```
s\dc("j1,1,0,4:r1,1,0,3:e2,1,2,5:r2,0,2,2"):approx(ir2)
```

This gives us a value for I_2 of **3.4** A. This is correct.

Bo2's Example 2.5 (Conductances)



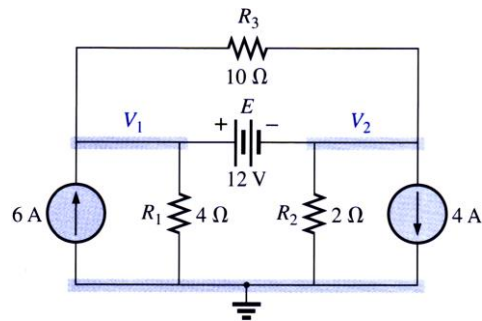
Determine the voltages in the nodes, and the current through the voltage source.

```
s\dc("j,0,1,3:e,3,2,3:r12,1,2,1/7:r20,2,0,1/3:r30,3,0,1/5:r13,1,3,1/2"):approx({v1,v2,v3,-ie})
```

The answer, **{-5,-1.5,1.5,11.5}**, is correct: $v_1=-.5$, $v_2=-1.5$, $v_3=1.5$, $i=11.5$

B11's Example 8.22

Determine V_1 and V_2 . My solution below:

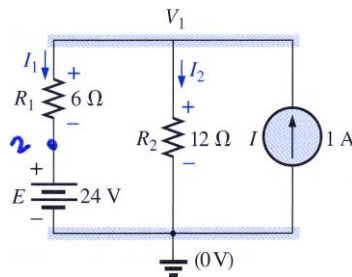


```
s\dc("j1,0,1,6:r1,1,0,4:e,1,2,12:r3,1,2,10:
r2,2,0,2:j2,2,0,4"):approx({v1,v2})
```

The answer, **{10.67,-1.33}**, tells us that V_1 is 10.67V and V_2 is -1.33V.

B11's Example 8.19

Determine V_1 , I_1 and I_2 .

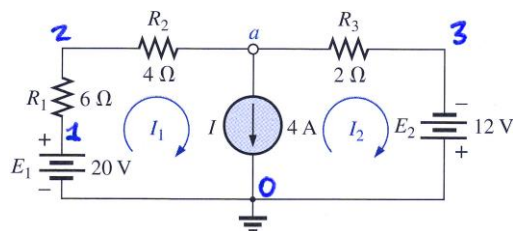


My solution below:

```
s\dc("e,2,0,24:r1,1,2,6:r2,1,0,12:j,0,1,1"):approx({v1,ir1,ir2})
```

The answer, **{20,-.667,1.67}**, tells us that V_1 is 20V, I_1 is -.667A and I_2 is 1.67V.

B11's Example 8.14



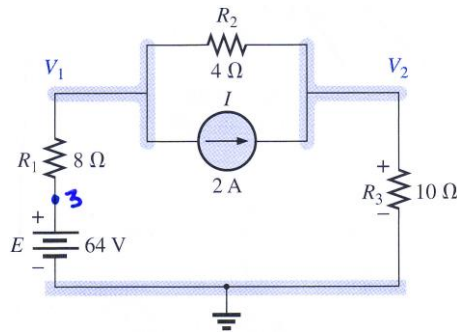
Determine I_2 and I_3 .

```
s\dc("e1,1,0,20:r1,1,2,6:r2,2,a,4:j,a,0,4:
r3,a,3,2:e2,0,3,12"):approx({ir2,ir3})
```

Answer: **{3.33,-.666}**. This is correct.

B11's Example 8.20

Determine V_1 , V_2 , I_1 , I_2 and I_3 . My solution is found below the circuit schematic:

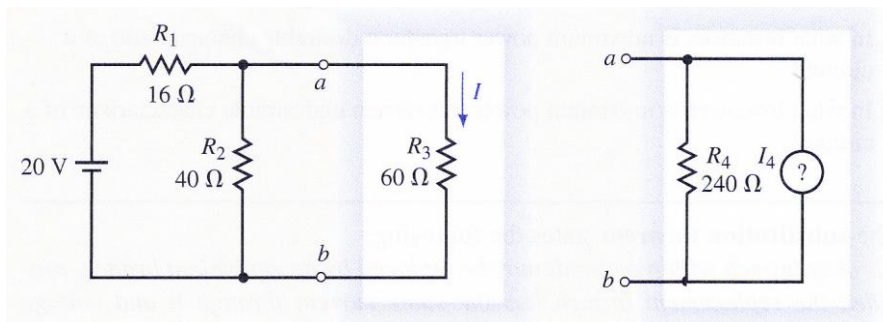


```
s\dc("e,3,0,64:r1,3,1,8:r2,1,2,4:j,1,2,2:r3,2,0,10"):approx({v1,v2,ir1,ir2,ir3})
```

Answer: **{37.82,32.73,3.27,1.27,3.27}**. You should know how to read these by now, but here it is just in case: $V_1 = 37.82\text{V}$, $V_2 = 32.73\text{V}$, $I_1 = 3.27\text{A}$, $I_2 = 1.27\text{A}$, $I_3 = 3.27\text{A}$.

RM3's Example 9-12 (solve)

If R_3 is to be replaced with R_4 and I_4 , determine the value and direction of the source.



First, make sure you understand what this problem is asking you to do. The idea is that, despite the change, we keep the same voltage drop and current flow between nodes **a** and **b**. We must first know what they are. So we simulate the original circuit:

```
s\dc("e,0,b,20:r1,0,a,16:r2,a,b,40:r3,a,b,60"): {va-vb,ir3}
```

We find that the voltage drop is 12V and the current is 0.2A. These are the currents and voltages that we have to keep once we do the replacement. Simulate the circuit now replacing R_3 with a resistor R_4 of 240Ω and a source j with value I_4 . Run this:

```
s\dc("e,0,b,20:r1,0,a,16:r2,a,b,40:r4,a,b,240:j,a,b,i4")
```

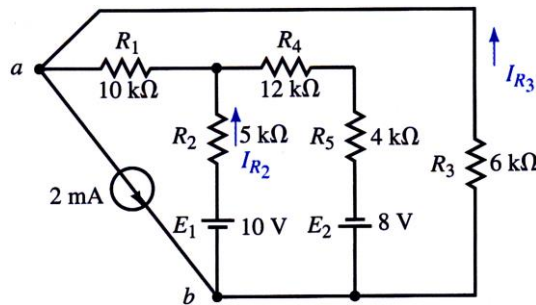
Notice that both the voltage drop (given by $va-vb$) and the current (given by $ir4+ij$) are algebraic functions in terms of $i4$. Now you can find $i4$ solving by voltage drop:

```
solve(va-vb=12.,i4) ...or by current flow... solve(ir4+ij=0.2,i4)
```

The result is the same: $i4 = .15\text{ A}$ The required current source is .15A from **a** to **b**.

RM3's Example 8-13

Solve for the currents through R_2 and R_3 in the circuit shown. My solution is below:

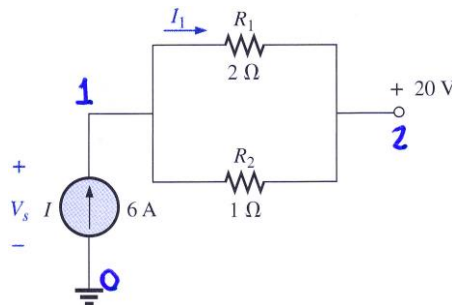


```
true→s\si:s\dc("r1,a,0,10k:r2,1,0,5k:r3,b,a,6k:r4,0,2,12k:
j,a,b,2m:e1,1,b,10:e2,b,2,8"):approx({ir2,ir3})
```

The answer, **{.00154,.00111}** is correct: $I_{R2} = 1.54$ mA and $I_{R3} = 1.11$ mA.

B11's Example 6.3 (Hidden source)

Determine V_s and I_1 .



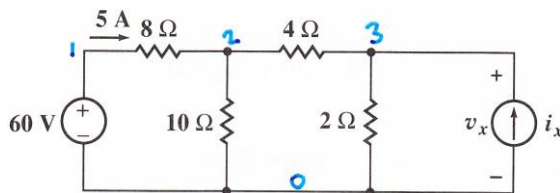
The problem presents a seemingly 'hanging' node with 20V. To simulate this, imagine a 'hidden' 20V voltage source connected to the node.

```
s\dc("j,0,1,6:r1,1,2,2:r2,1,2,1:e,2,0,20"): {v1,ir1}
```

We get the answer: **{24,2}**, which is correct. V_s is **24V** and I_1 is **2A**.

HK5's Figure 1-24c (Expert)

Determine i_x and v_x in the following circuit.

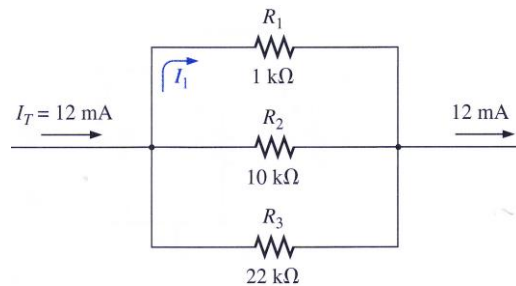


```
s\ex("e,1,0,60:r8,1,2,8:r10,2,0,10:r4,2,3,4:r2,3,0,2:j,0,3,ix"): {ix,v3}
```

Select DC. Add unknown i_x . Add equation $i_{R8}=5$. Run the simulation, and you will get: **{1,8}**. This is correct: I_x is 1A and that V_x is 8V.

B11's Example 6.22 (Hidden source)

Determine I_1 .



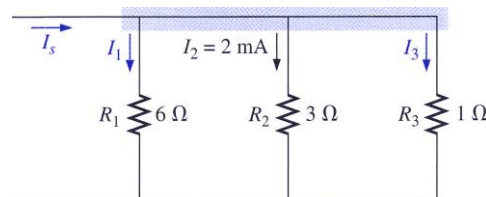
Although no source is shown, the circuit has a current. I use a 'hidden' current source.

```
true→s\si:s\dc("jt,0,1,12m:r1,1,0,1k:r2,1,0,10k:r3,1,0,22k"):approx(ir1)
```

We get $I_1 = 10.48 \text{ mA}$, which is correct.

B11's Example 6.21 (Hidden source, Expert)

Determine I_s , I_1 and I_3 .



This 'hidden source' problem is perfect for Expert. My solution:

```
true→s\si:s\ex("js,0,1,is:r1,1,0,6:r2,1,0,3:r3,1,0,1"):approx({is,ir1,ir3})
```

Select DC, and press Enter. Add unknown I_s . Add equation $ir2=2m$ Run the simulation. The answer, $\{.009,.001,.006\}$, is correct, since the currents are as follows: I_s is 9mA, I_1 is 1mA and I_3 is 6mA.

Resistors in parallel

Symbulator has a way to help you reduce two resistors connected in parallel to their equivalent. It works as part of circuit descriptions, and is typed thus: $p(,)$. When you do not need to know the current through or power consumed in each individual resistor, you can reduce parallel resistors using $p(,)$ inside your circuit description.

Examples

As a first example, here is a simulation in which it makes sense to use $p(,)$. In B11's Example 7.4 (shown in page 116), it makes sense to reduce R_4 and R_5 to an equivalent resistor, since we do not need to know their individual currents or power use.

```
s\dc("e,1,0,16.8:r1,1,2,9:r2,1,2,6:r3,2,3,4:re,3,0,p(6,3):r6,2,0,3")
```



...provides the right answers, e.g. I_B is $I_{R3} = 1$ A and voltage drop in area B is $V_2 = 6$ V.

The second example is of a simulation in which using **p()** makes no sense. In B11's Example 8.3 (shown in page 130), it makes no sense to reduce R_1 and R_2 , because you need to know the value of the current through R_1 .

As third example, here is a simulation where you can reduce part of the resistors using **p()**. In B11's Example 6.22 (shown in page 130), we must leave R_1 alone, because we need to know the current through it, but we can reduce R_2 and R_3 to their equivalent using **p()**, since we do not need to know their individual currents or power use:

```
true→s\si:s\dc("jt,0,1,12m:r1,1,0,1k:re,1,0,p(10k,22k)")
```

...provides the same answer we got before, e.g. $I_1 = 10.48$ mA.

Moving forward, we will use the **p()** tool whenever we feel it is appropriate. A similar reduction of resistors in series is possible through simple addition whenever we do not need to know specific answers for each resistor, such as the voltage drop or power consumed in each, or the voltage in the node between them. The current through series resistors is the same, so even through an equivalent you can get the current.

Dependent sources

One of my favorite scenes in cinema comes from *The Dark Knight*: the Joker (played by Heath Ledger) is rolling on the floor of a Gotham City prison, taking a bare-knuckle beating from an ever-more-frustrated Batman. Master of the situation and laughing hysterically, the Joker says: "*You have nothing! Nothing to threaten me with!*"

Even though the movie had not been made yet, I remember feeling something along the same lines – although maybe less hysteric – back in 1999 when I realized that one of the consequences of having used a 100% symbolic implementation for Symbulator meant that I could make any element's value dependent on any answer of the circuit. I could simulate voltage or current sources that were dependent on any voltage, current or combination thereof, with the same ease that I could simulate a 12V source.

**For Symbulator,
dependent
sources are just
sources. They
require no
special notation.**

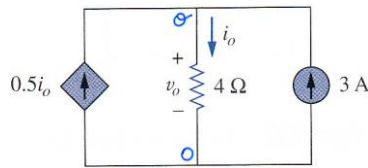
Here is what you need to know for simulating dependent sources on Symbulator: *nothing*. There is nothing special to it, nothing at all. Just write the value as a function of the circuit's answers, using the variables that by now you should know well, and run the simulation like it's nobody's business. With Symbulator, instead of fearing them, you will laugh in the face of dependent sources, thinking: "*You have nothing!*" Booyah!

Without any more introduction, let me show you examples, so you can learn by doing.

Numerical examples

AS2's Example 2.7

Find i_o and v_o in the circuit below.



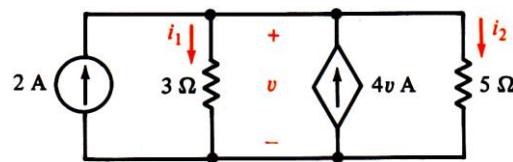
My solution below. I named the top node **o** (the letter) and used the bottom node as ground so that when I ask for **vo** I get the voltage in node o, matching the book's variable v_o . Likewise I name the resistor **ro**, so that I can ask for **iro** to get current i_o . None of this special labeling is necessary, but it's kind of cool when the variable Symbulator gives you is called the same as the one the book is asking for. Chalk it up to my nerdiness. Notice that **5iro** is interpreted by the calculator as 5 times **iro**.

```
s\dc("ji,0,o,3:ro,o,0,4:jd,0,o,.5iro"): {iro,vo}
```

The answer, **{6.,24.}**, is correct: $i_o = 6$ and $v_o = 24$.

Bo2's Example 1.9

Determine i_1 , i_2 and v .



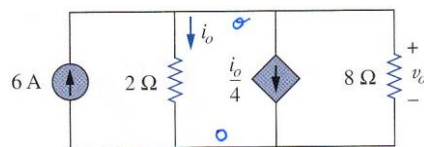
This is my solution. The simulation took 10 seconds.

```
s\dc("ji,0,1,2:r1,1,0,3:jd,0,1,4v1:r2,1,0,5"): {ir1,v1,ir2}
```

The answer, **{-5/26,-15/26,-3/26}**, is correct: $i_1 = -5/26$, $i_2 = -3/26$ and $v = -15/26$.

AS2's Practice Problem 2.7

Find v_o and i_o in the circuit. My solution below:

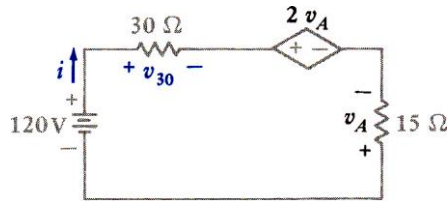


```
s\dc("ji,0,o,6:ro,o,0,2:jd,o,0,iro/4:r8,o,0,8"): {vo,iro}
```

The answer, **{8,4}**, is correct: $v_o = 8$ and $i_o = 4$.

HK5's Example 1-3

Determine the power delivered by each source and consumed by both resistors.

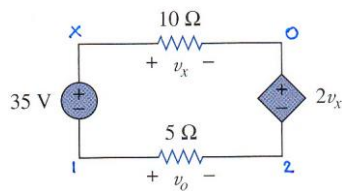


`s\dc("ei,1,0,120:r1,1,2,30:ed,2,3,2vra:ra,0,3,15"):{-pei,-ped,pr1+pra}`

The answer, **{960,1920,2880}**, is right: the independent source delivers 960W, the dependent source delivers 1920W, and the resistors consume 2880W together.

AS2's Practice Problem P2.6

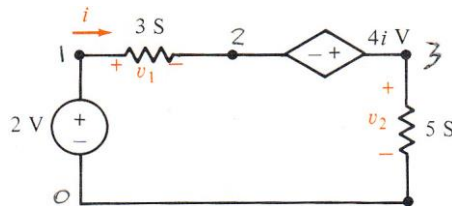
Find v_x and v_o in the circuit. My solution below:



`s\dc("ei,x,1,35:rx,x,0,10:ed,0,2,2vx:ro,1,2,5"):{vx,vro}`

The answer, **{10,-5}**, is correct: $v_x = 10$ and $v_o = -5$.

Bo2's Example 1.10

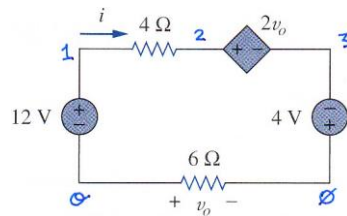


Determine v_1 , v_2 and i . My solution below. The simulation took 14 seconds.

`s\dc("ei,1,0,2:r1,1,2,1/3:ed,3,2,4*ir1:r2,3,0,1/5"):{ir1,vr1,vr2}`

The answers, **{-15/26,-5/26,-3/26}**, is correct: $v_1 = -5/26$, $v_2 = -3/26$ and $i = -15/26$.

AS2's Example 2.6



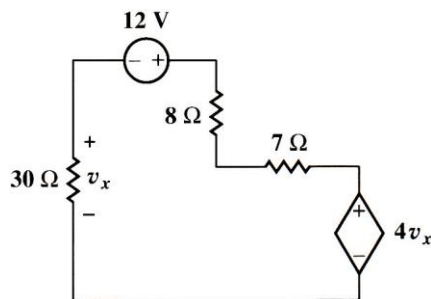
Determine v_o and i in the circuit. My solution is shown below:

`s\dc("e12,1,0,12:ri,1,2,4:ed,2,3,2vo:e4,0,3,4:ro,0,0,6"):{vo,iri}`

The answer, **{48,-8}**, is correct: $v_o = 48$ and $i = -8$.

HK5's Drill Problem 1.11

Find the power absorbed by each element in the circuit. My solution below:

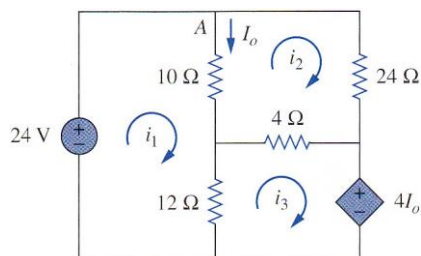


`s\dc("r1,x,0,30:ei,1,x,12:r2,1,2,8:r3,2,3,7:ed,3,0,4vx"):approx({pr1,pei,pr2,pr3,ped})`

The answer, **{.768,1.92,.2048,.1792,-3.072}**, is correct.

AS2's Example 3.6

Determine the value of I_o in the circuit. My solution below:

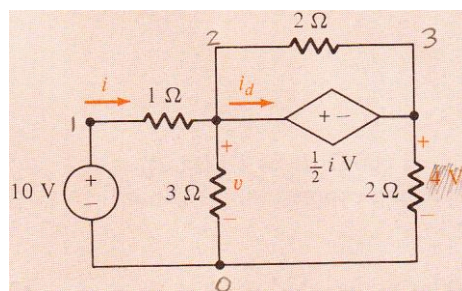


`s\dc("ei,a,0,24:ro,a,b,10:r12,b,0,12:r4,b,c,4:r24,a,c,24:ed,c,0,4iro"):approx(iro)`

The answer, **1.5 A**, is correct.

Bo2's Drill Exercise 1.12

Determine i , v and i_d .



We are given an unnecessary piece of information: the 4V drop in the 2Ω resistor.

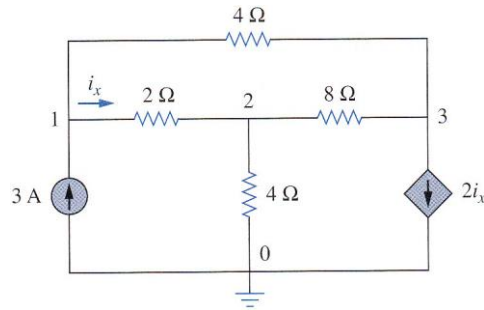
My solution is shown below the schematic. My simulation took 18 seconds.

```
s\dc("ei,1,0,10:r1,1,2,1:r2,2,3,2:r3,2,0,3:
r4,3,0,2:ed,2,3,ir1/2"): {ir1,vr3,ied}
```

The answer, **{4,6,1}**, is correct: $i=4$, $v=6$ and $i_d=1$.

AS2's Example 3.2

Determine the voltages at the nodes.



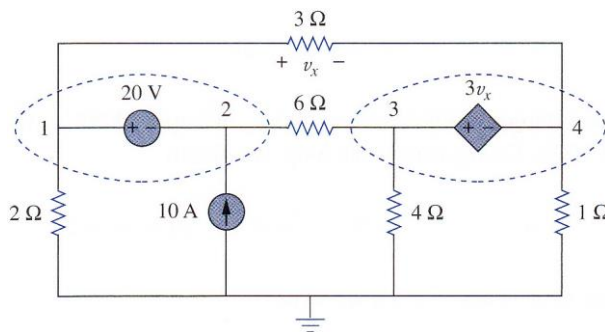
My solution below:

```
s\dc("ji,0,1,3:jd,3,0,2ir2:r2,1,2,2:r4a,1,3,4:r8,2,3,8:r4b,2,0,4"):approx({v1,v2,v3})
```

The answer, **{4.8,2.4,-2.4}**, is correct.

AS2's Example 3.4

Find the node voltages in the circuit. My solution below:



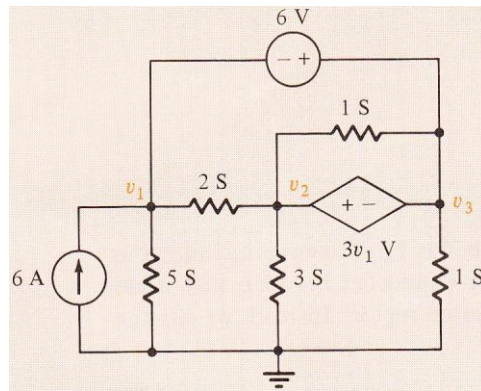
```
s\dc("r2,1,0,2:e,1,2,20:j,0,2,10:r6,2,3,6:rx,1,4,3:r4,3,0,4:
ed,3,4,3vr:r1,4,0,1"):approx({v1,v2,v3,v4})
```

The answer, **{26.67,6.67,173.33,-46.67}**, is correct.

Bo2's Drill Exercise 2.6

Determine the voltage in each node.

Notice that in the schematic, the resistors' values are given in siemens. Symbolator has to be fed the resistors with values in ohms. This means that, when we simulate them, we have to convert them from siemens to ohms by dividing 1 over the siemens value.



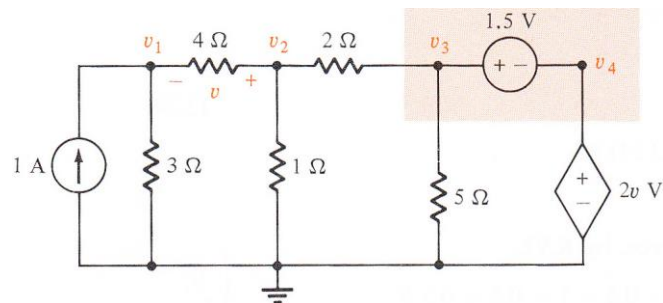
My solution below:

```
s\dc("j,0,1,6:ei,3,1,6:ed,2,3,3v1:r5,1,0,1/5:r2,1,2,1/2:
r3,2,0,1/3:r1,2,3,1:r4,3,0,1"): {v1,v2,v3}
```

The answer, **$\{-1, 2, 5\}$** , is correct.

Bo2's Example 2.7

Determine the voltages in all nodes. My solution below the schematic:

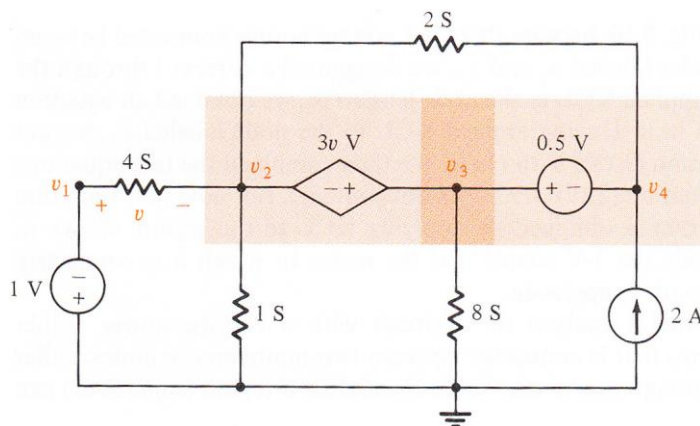


```
s\dc("j,0,1,1:r3,0,1,3:r4,2,1,4:r1,2,0,1:r2,2,3,2:
r5,3,0,5:ei,3,4,1.5:ed,4,0,2vr4"): {v1,v2,v3,v4}
```

The answer, **$\{1.5, -5, -2.5, -4\}$** , is correct.

Bo2's Example 2.6

Determine the voltages in all nodes.



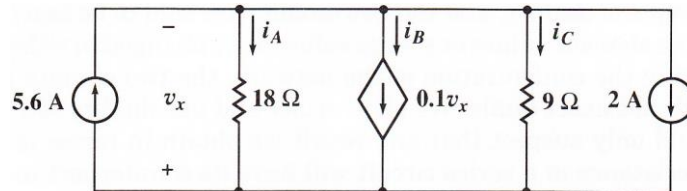
My solution below:

```
s\dc("e1,0,1,1:e2,3,4,.5:ed,3,2,3vr4:j,0,4,2:r4,1,2,1/4:
r1,2,0,1:r8,3,0,1/8:r2,2,4,1/2"):v1,v2,v3,v4}
```

The answer, **{-1,-2,1.,.5}**, is correct.

HK5's Drill Problem 1-12

Find i_A , i_B and i_C .



My solution below:

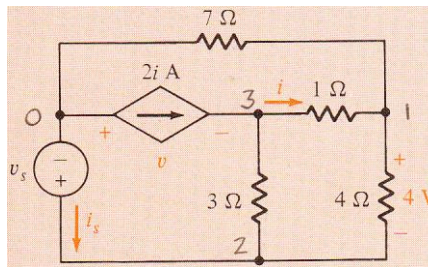
```
s\dc("jl,x,0,5.6:ra,0,x,18:jb,0,x,.1vx:r9,0,x,9:
jr,0,x,2"):approx({ira,ijb,ir9})
```

The answer, **{3.,-5.4,6.}**, is correct.

Numerical-from-symbolic examples

Bo2's Drill Exercise 1.10

Determine i , v , i_s and v_s .



With one unknown value and one known solution, this problem is a job for Expert.

Determine i , v , i_s and v_s .

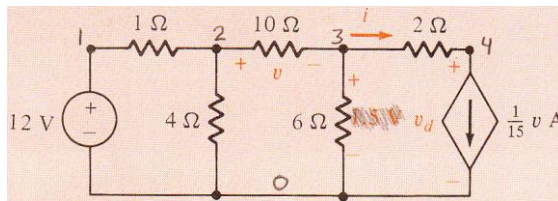
```
s\ex("es,2,0,vs:jd,0,3,2ir1:r7,0,1,7:r1,3,1,1:
r3,3,2,3:r4,1,2,4"):v1,vjd,-ies,vs}
```

Add `,vs` to the unknowns, and add `and vr4=4` to the equations. Run the simulation.

The answer, **{2,-9,-3,3}**, is correct: $i=2$, $v=-9$, $i_s=-3$ and $v_s=3$.

Bo2's Drill Exercise 1.11

Determine i , v and v_d . (Since all element values are known, the tip we are given by the book – namely, that the voltage drop in the 6Ω resistor is $1.5V$ – is totally superfluous.)



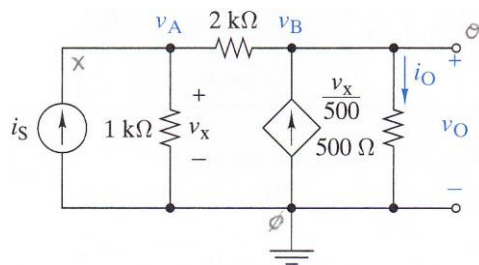
```
s\dc("e,1,0,12:r1,1,2,1:r4,2,0,4:r10,2,3,10:r6,3,0,6:
r2,3,4,2:j,4,0,vr10/15"):approx({vr10,ir2,vj})
```

The answer, **{7.5,.5,.5}** is correct: $i=.5$, $v=7.5$ and $v_d=.5$.

Symbolic examples

TR5's Exercise 4.2

Find v_o and i_o in terms of i_s .



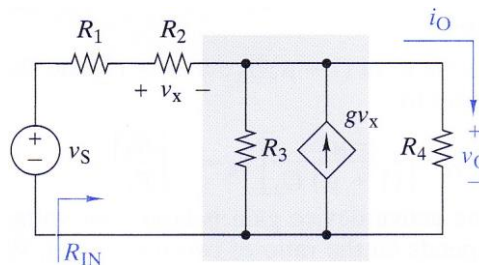
This is my solution. The simulation took 14 seconds.

```
true→s\si:s\dc("ji,0,x,is:r1,x,0,1k:r2,x,0,2k:jd,0,o,vx/500:ro,o,0,500"):
{vo,iro}
```

The answer, **{1000*i_s,2*i_s}**, is correct: $v_o=1000 i_s$ and $i_o=2 i_s$.

TR5's Example 4.4

Find v_o and the equivalent resistance R_{IN} , in terms of v_s , when R_1 is 50, R_2 is 1k, R_3 is 100, R_4 is 5k and g is 100mA (e.g. 100m).



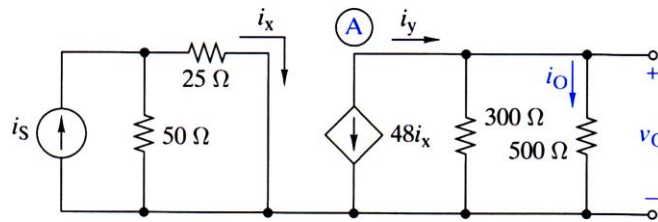
This is my solution. The simulation took 18 seconds.

```
true→s\si:s\dc("e,1,0,vs:r1,1,2,50:r2,2,o,1k:r3,o,0,100:
r4,o,0,5k:j,0,o,100m*vr2"):approx({vo,re})
```

The answer, **{.904*v_s,10951.}**, is correct: $v_o=.904 v_s$ and $R_{IN}=10.95K\Omega$.

TR5's Figure 4-4

Find the voltage drop, current, and power consumed by the 500Ω resistor, and the ratio of that power to that delivered by the independent source, all in terms of i_s .



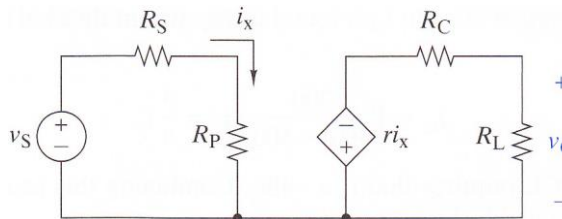
I have not labeled the nodes in the figure, so you can practice doing it. My solution:

```
s\dc("js,0,1,is:r50,1,0,50:rx,1,0,25:jd,o,0,48irx:
r3,o,0,300:ro,o,0,500"):{iro,vo,pro,pro/(-pjs)}
```

The answers we get are correct: $i_o = -12i_s$, $v_o = -6000i_s$, $p_o = 72000i_s^2$, and $p_o/p_s = 4320$.

TR5's Example 4.1 (Symbolic)

Find v_o .



I did not label the nodes, so you can practice. Remember not using **rc**: it is reserved.

```
s\dc("ei,1,0,vs:rs,1,2,rs:rx,2,0,rp:ed,0,3,r*irx:rrc,3,o,rrc:rl,o,0,rl"):vo
```

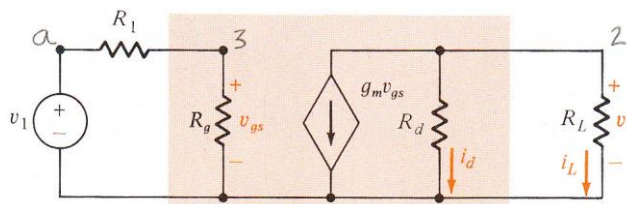
The answer, shown left, is correct. The textbook's answer is shown right.

$$\frac{-r_l \cdot v_s \cdot r}{(r_l + r_{rc}) \cdot (r_p + r_s)}$$

$$v_o = \left[\frac{-R_L r}{(R_s + R_p)(R_c + R_L)} \right] v_s$$

Bo2's Example 1.11 (Symbolic)

Determine v_2 .



My solution below. In my solution I named the value of the source v_1 , to keep it similar to the book. This required avoiding naming any node as **1**: if there was a node 1,

Symulator would store in **v1** the voltage of the node, creating trouble. There is no problem with using r1 as a value, since nothing will be stored in that r1 value.

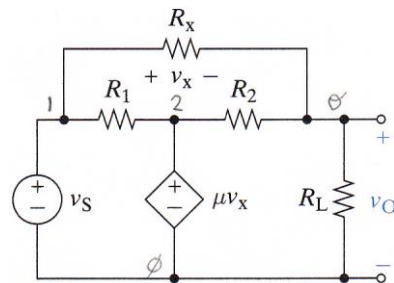
```
sldc("e,a,0,v1:r1,a,3,r1:rg,3,0,rg:j,2,0,gm*vrg:
rd,2,0,rd:rl,2,0,rl"):v2
```

The simulation took 25 seconds. The answer I got is shown left (the textbook's right.)

$$\frac{-g_m \cdot r_d \cdot r_g \cdot r_l \cdot v_1}{(r_1 + r_g) \cdot (r_d + r_l)} \quad v_2 = \frac{-g_m R_d R_g R_L v_1}{(R_g + R_1)(R_L + R_d)}$$

TR5's Exercise 4.3 (Symbolic)

Find v_o , in terms of the value in the circuit. For resistors, use their conductance value.



This is my solution. Since we want to use μ as a constant in the dependent source, set the `s\si` flag to false, so that it is not confused with the SI unit for micro.

```
false→s\si:sldc("ei,1,0,vs:ed,2,0,μ*(vr_x):r1,1,2,1/g1:
r2,2,0,1/g2:rx,1,0,1/gx:rl,0,0,1/gl"):vo
```

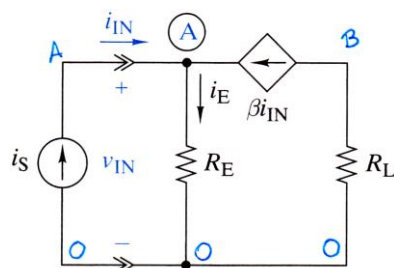
This is the answer we get. It is correct. Compare it with the textbook's answer.

$$\frac{(g_2 \cdot \mu + g_x) \cdot v_s}{g_2 \cdot (\mu + 1) + g_l + g_x} \quad v_o = \frac{G_x + \mu G_2}{G_x + G_L + (\mu + 1)G_2} v_s$$

The last four problems show Symulator at its DC best. I don't know of any calculator-based program that was able to provide this kind of purely symbolic answer to a circuit simulator back in 1999 when I made Symulator. As a matter of fact, even today – fifteen years later - I know of no other calculator-based simulator that can do this.

TR5 Example 4-7 (Symbolic)

Find R_{IN} , e.g. the resistance as seen by the current source. My solution below.





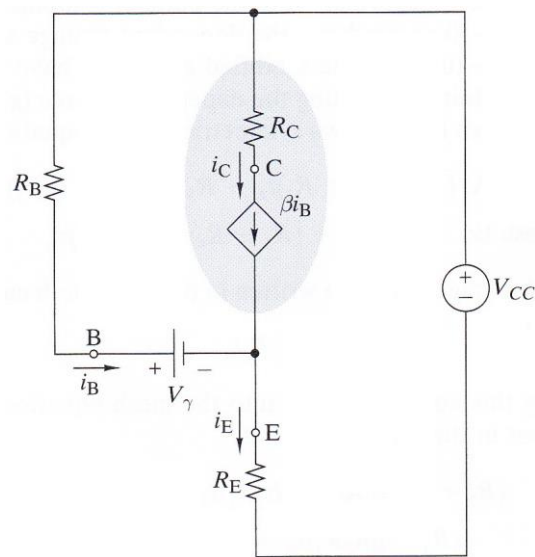
$$R_{IN} = (\beta + 1)R_E$$

The answer we get, βR_E , is correct, as can be seen by comparing it to the textbook's answer, shown below.

$$R_{IN} = (\beta + 1)R_E$$

TR5's Example 4.5 (Symbolic)

Find i_B .



My solution is shown below. Be patient. This simulation takes more than one minute.

$$i_B = \frac{V_{CC} - V_{\gamma}}{R_B + (\beta + 1)R_E}$$

Compare my answer, left, to the book's answer, right.

$$i_B = \frac{V_{CC} - V_{\gamma}}{R_B + (\beta + 1)R_E}$$

Element: s

Short circuits

Describing an ideal short circuit

Shorts are used mostly when we need to find out a current in a part of the circuit where there is no element already. Otherwise, we would just define it as a single node. In Symbolator, an ideal **short circuit** is described as follows: first the name of the short, which must start with the letter **s**, and a coma; second the name of the first node, another coma; and third the name of the second node. There is no coma at the end.

This is how short circuits are described in Symbolator.

For example, a short circuit called **s1**, connected between nodes **3** and **5** (e.g. current flowing through it from node 3 towards node 5), would be described thus: **s1,3,5**

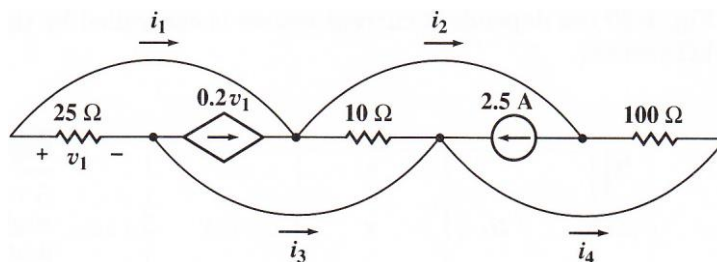
Simulation answers

No power is consumed or voltage is dropped in a short. For each short in a circuit, Symbolator stores only the *current through it*, flowing from the first node towards the second, in a variable called **i** and the name of the short. For example, the current flowing through a short called **sx**, from the first to second node, is stored in **isx**.

An example using short circuits

HK5's Drill Problem 1-13

Find i_1 , i_2 , i_3 and i_4 .



My solution below. I define the shorts in the same direction as the arrows.

```
s\dc("r1,1,0,25:jd,0,2,.2v1:r2,2,3,10:ji,4,3,2.5:r3,4,5,100:
s1,1,2:s2,2,4:s3,0,3:s4,3,5"):approx({is1,is2,is3,is4})
```

The answer, **$\{-2., 3., -8., -5.\}$** , is correct.

This answer can only be found in Symbolator using short circuits.

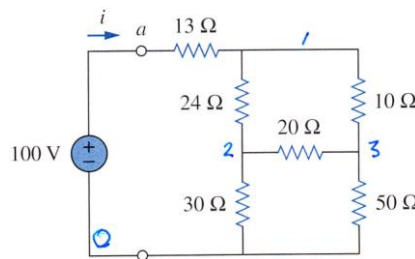
Equivalents

Equivalent resistance

As we saw before, Symbolator has the ability to provide the equivalent resistance of a circuit as seen from any source. This allows us to solve problems like the following:

AS2's Practice Problem 2.15

Find the equivalent resistance as seen by the 100V source, and the value of current i .



```
sldc("e,a,0,100:r13,a,1,13:r24,1,2,24:r10,1,3,10:
r20,2,3,20:r30,2,0,30:r50,3,0,50")
```

By evaluating `re` we get the value of the equivalent resistance as seen by the source `e`. It is 40Ω. By evaluating `ir13` we get a value of 2.5A for current i . The answers were easily found, because there was a source connected between the two desired nodes.

But how can we find the equivalent resistance of a passive circuit, that is to say, a circuit that has no independent source in it? One way to find this in Symbolator is to connect a 1A current source between the two nodes where we want to find the equivalent resistance, and then read the voltage drop across the source.

An easier way is to run the `er` script, which does the same thing for you, automatically.

Script: `er`

*The `er` script
finds the
equivalent
resistance of a
circuit, and
stores it in `req`*

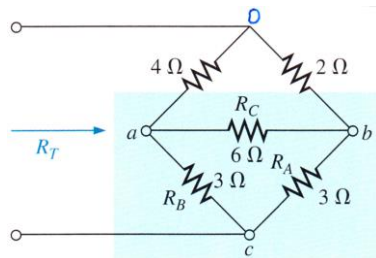
The `er` script finds the equivalent resistance of a passive circuit, and stores it – in the case of a DC analysis – in a variable called `req`. The `er` script takes three arguments: the first is the circuit description in the form of a string; the second and third are the two nodes between which we need to find the equivalent resistance. Let's learn by doing.

Example for a resistive circuit

B11's Example 8.29

Calculate the equivalent resistance of the circuit shown below.

Let me solve this problem step by step, since it is the first time you see the **er** script.



After I label the nodes, I describe the circuit. In this case, I store it in a variable.

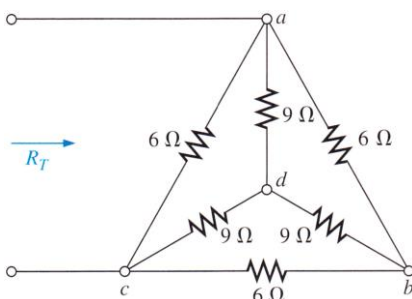
```
"r4,0,a,4:r2,0,b,2:r6,a,b,6:rb,a,c,3:ra,b,c,3"→cir
```

Run the **er** script, giving it as arguments the circuit and the nodes: `s\er(cir,0,c)` When prompted, choose DC as analysis type. Once it is done, evaluate `approx(req)` The value is **2.89** Ω. This is correct. Below are many practice examples of this type.

Practice problems for resistive circuits

B11's Example 8.30

Find the equivalent resistance of the circuit below.

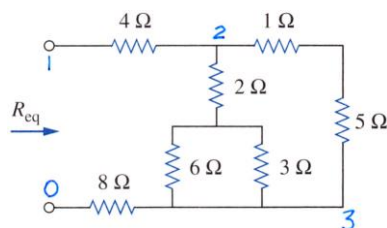


```
s\er("rac,a,c,6:rad,a,d,9:rab,a,b,6:rcd,c,d,9:rbc,b,c,6:rbd,b,d,9",a,c)
```

Choose DC. When *Done*, use `approx(req)` to find the equivalent resistance is **3.27** Ω.

AS2's Example 2.9

Find the equivalent resistance of the circuit below.



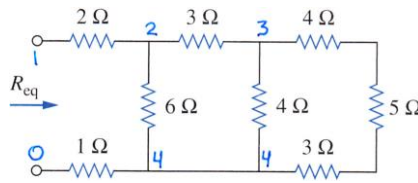
Please notice my use of **p()** to reduce resistors in parallel, and **+** for resistors in series.

```
s\er("r12,1,2,4:r03,0,3,8:r23,2,3,p(1+5,2+p(6,3))",1,0)
```

Choose DC. When *Done*, use `approx(req)` to find the equivalent resistance is **14.4 Ω** .

AS2's Practice Problem 2.9

Find the equivalent resistance of the circuit below.



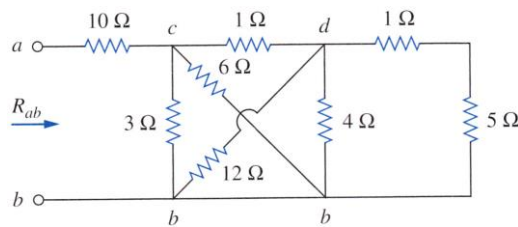
In my circuit description below, please notice the use of **p()** to reduce resistors in parallel, and **+** to reduce resistors in series.

```
s\er("r12,1,2,2:r04,0,4,1:r24,2,4,6:r23,2,3,3:r34,3,4,p(4,4+5+3)",1,0)
```

Choose DC. When *Done*, evaluate `req` to find the equivalent resistance is **6 Ω** .

AS2's Example 2.10

Find the equivalent resistance of the circuit below.



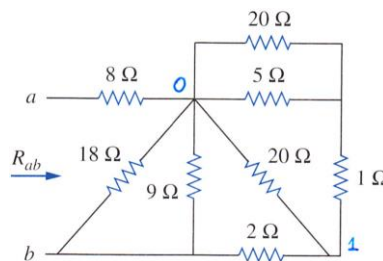
In my circuit description below, please notice the use of **p()** to reduce resistors in parallel, and addition to reduce resistors in series.

```
s\er("rac,a,c,10:rbc,b,c,p(3,6):rcd,c,d,1:rbd,b,d,p(p(12,4),1+5)",a,b)
```

Choose DC. Wait for *Done*. Evaluate `approx(req)`. The equivalent resistance is **11.2 Ω** .

AS2's Practice Problem 2.10

Find the equivalent resistance of the circuit below.

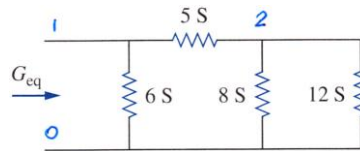


```
s\er("ra0,a,0,8:rb0,b,0,p(18,9):rb1,b,1,2:r10,1,0,p(20,1+p(20,5))",a,b)
```

Choose DC. When *Done*, evaluate `req`, and find the equivalent resistance is **11 Ω** .

AS2's Example 2.11

Find the equivalent *conductance* of the circuit below.

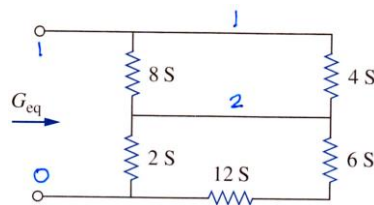


`s\er("r12,1,2,1/5:r10,1,0,1/6:r20,2,0,p(1/8,1/12)",1,0)`

Choose DC. Wait for *Done*. Evaluate $1/r_{eq}$ to find the equivalent *conductance* is **10 S**.

AS2's Practice Problem 2.11

Find the equivalent *conductance* of the circuit below.



`s\er("r12,1,2,p(1/8,1/4):r20,2,0,p(1/2,1/12+1/6)",1,0)`

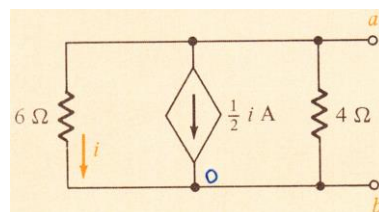
Choose DC. Wait for *Done*. Evaluate $1/r_{eq}$ to find the equivalent *conductance* is **4 S**.

Examples with dependent sources

The **er** script can be applied to a second type of passive circuits: circuit that include resistors and dependent sources, but no independent sources. A circuit with no independent sources can only be reduced to an equivalent resistance, not to a Thévenin or Norton equivalent, so the proper script to use is **er**. The script is used in exactly the same manner as for resistive circuits, making sure the dependent sources are described properly, according to the conventions of Symbolator notation. Below is a series of examples of passive circuits with dependent sources.

Bo2's Drill Problem 3.14

Find the equivalent resistance of the circuit below.

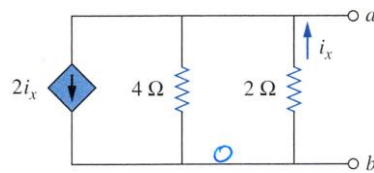


`s\er("r4,a,0,4:ri,a,0,6:ji,a,0,iri/2",a,0)`

Choose DC. Wait for *Done*. Evaluate r_{eq} to find the equivalent resistance is **2 Ω**.

AS2's Example 4.10

Find the equivalent resistance of the circuit below.

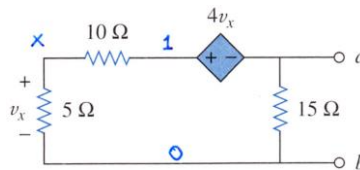


`sler("r4,a,0,4:rx,0,a,2:j,a,0,2irx",a,0)`

Choose DC. Wait for *Done*. Evaluate `req`. The equivalent resistance is **-4 Ω**. It may be surprising to have a negative resistance. This is the result of the dependent sources.

AS2's Practice Problem 4.10

Find the equivalent resistance of the circuit below.

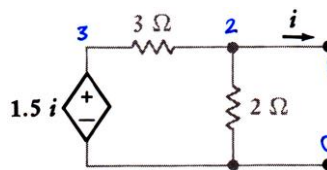


`sler("r15,a,0,15:e,1,a,4vr:x:r10,1,x,10:rx,x,0,5",a,0)`

Choose DC. Wait for *Done*. Evaluate `req`, we find the equivalent resistance is **-7.5 Ω**.

HK5's Figure 2-29

Find the equivalent resistance of the circuit below.

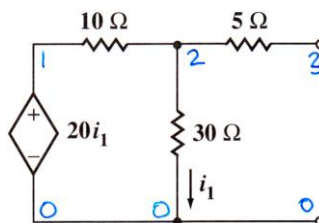


`sler("e,3,0,1.5is:r3,3,2,3:r2,2,0,2:s,2,1",1,0)`

Choose DC. Wait for *Done*. Evaluate `req`. The equivalent resistance is **0.6 Ω**.

HK5 Drill Problem 2-9d

Find the equivalent resistance of the circuit below.

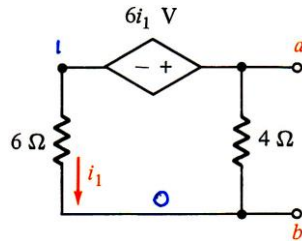


```
s\er("r10,1,2,10:r5,2,3,5:r1,2,0,30:e,1,0,20ir1",3,0)
```

Choose DC. Wait for *Done*. Evaluate `req`. The equivalent resistance is **20** Ω .

Bo2's Example 3.12

Find the equivalent resistance of the circuit below.



```
s\er("r1,1,0,6:r4,a,0,4:e,a,1,6ir1",a,0)
```

Choose DC. Wait for *Done*. Evaluate `req`. The equivalent resistance is **3** Ω .

Thévenin/Norton

Just like a passive circuit can be reduced to an equivalent resistance, an active circuit (e.g. one with independent sources) can be reduced to a Thévenin or Norton equivalent. One way to find the Thévenin or Norton equivalent using Symulator is to run a first simulation of the circuit to find out the voltage between the two nodes where we want to find the equivalent (this is the Thévenin voltage, or V_{TH}), and then to run a second simulation with a short-circuit connected between these two nodes to find out the circuit running through it (this is the Norton current, or I_{NO}). We can then get the equivalent resistance, or R_{EQ} , by dividing V_{TH}/I_{NO} . Notice that R_{EQ} is often called the Thévenin resistance.

An easier way to find the Thévenin and Norton equivalent of a circuit in Symulator is to run the **th** script, which does exactly the same thing described above, automatically.

Script: **th**

The **th** script finds the Thévenin and Norton equivalents of an active circuit. It takes three arguments: the circuit description, the first node and the second node. With **th**:

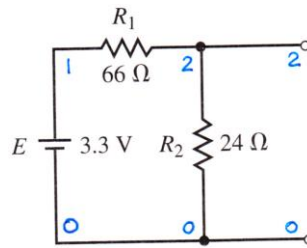
- the value of the Thévenin voltage, V_{TH} , is stored in **vth**
- the value of the Norton current, I_{NO} , is stored in **ino**, and
- the value of equivalent resistance, R_{EQ} , is stored in **req**.

th also stores other useful expressions into variables, which we will discuss later.

*The **th** script finds the Thevenin and Norton equivalents of a circuit. It stores the answers in variables as shown left.*

RM3's Practice Problem 9-4

Find the Thévenin and Norton equivalents of the circuit below.

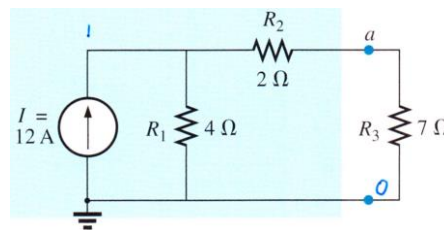


`s\th("e,1,0,3.3:r1,1,2,66:r2,2,0,24",2,0)`

Pick DC. Via `vth` find $V_{TH} = 0.88$ V. Via `ino`, find $I_{NO} = 0.05$ A. Via `req`, find $R_{EQ} = 17.6$ Ω.

B11's Example 9.7

Find the Thévenin equivalent of the circuit below, as seen from the R_3 resistor.

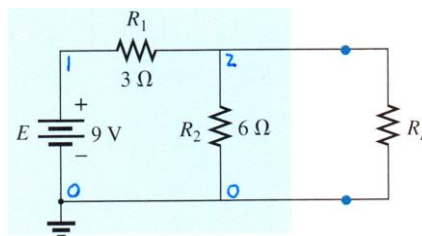


`s\th("j,0,1,12:r1,1,0,4:r2,1,a,2",a,0)`

Choose DC. Via `vth` we find $V_{TH} = 48$ V. Via `req` we find $R_{EQ} = 6$ Ω.

B11's Example 9.11

Find the Norton equivalent of the circuit below, as seen from the R_L resistor.

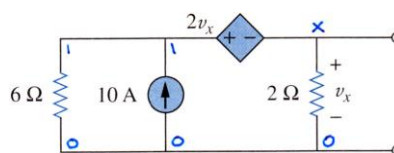


`s\th("e,1,0,9:r1,1,2,3:r2,2,0,6",2,0)`

Choose DC. Via `ino` we find $I_{NO} = 3$ A. Via `req` we find $R_{EQ} = 2$ Ω.

AS2's Practice Problem 4.12

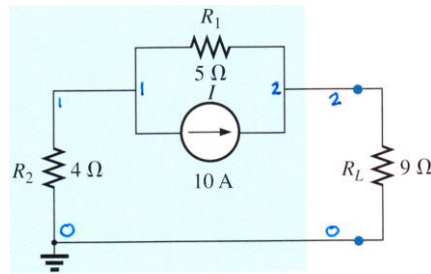
Find the Norton equivalent of the circuit below.



`s\th("r6,1,0,6:j,0,1,10:r2,x,0,2:e,1,x,2vx",x,0)`

Choose DC. Via `ino` we find $I_{NO} = 10$ A. Via `req` we find $R_{EQ} = 1\ \Omega$.

B11's Example 9.12



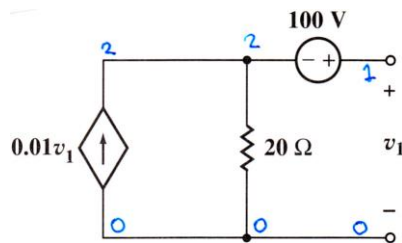
Find the Norton equivalent of the circuit, as seen from the R_L resistor.

`s\th("r2,1,0,4:r1,1,2,5:j,1,2,10",2,0)`

Choose DC. Via `ino` we find $I_{NO} = 5.56$ A. Via `req` we find $R_{EQ} = 9\ \Omega$.

HK5's Drill Problem 2-8b

Find the Thévenin equivalent of the circuit below.

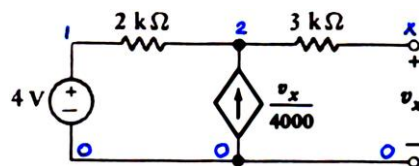


`s\th("j,0,2,0.01v1:r,0,2,20:e,1,2,100",1,0)`

Choose DC. Via `vth` we find $V_{TH} = 125$ V. Via `req` we find $R_{EQ} = 25\ \Omega$.

HK5's Figure 2-27

Find the Thévenin equivalent of the circuit below.



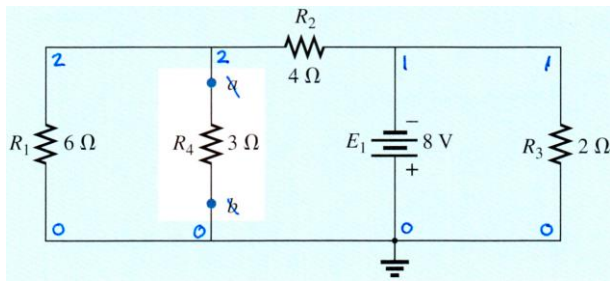
`true→s\si:s\th("e,1,0,4:r2,1,2,2k:r3,2,x,3k:j,0,2,vx/4000",x,0)`

Via `vth` we find $V_{TH} = 8$ V. Via `req` we find $R_{EQ} = 10$ kΩ.

B11's Example 9.8

Find the Thévenin equivalent of the circuit below, as seen from the R_4 resistor.

I ignore the textbook's decision to call the nodes **a** and **b**, since b is ground anyway.

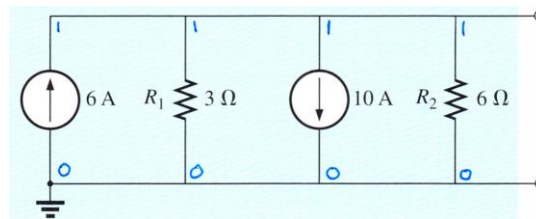


$\text{sth}("r1,2,0,6:r2,2,1,4:r3,1,0,2:e,0,1,8",2,0)$

Choose DC. Via **vth** we get $V_{TH} = -4.8$ V. Via **req** we get $R_{EQ} = 2.4$ Ω.

B11's Example 8.6

Find the Norton equivalent of the circuit below.

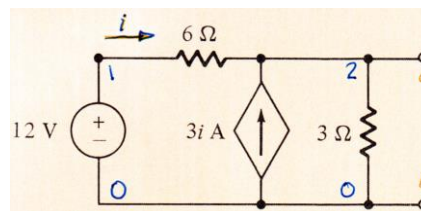


$\text{sth}("j1,0,1,6:r1,1,0,3:j2,1,0,10:r2,1,0,6",1,0)$

Choose DC. Via **ino** we find $I_{NO} = -4$ A. Via **req** we find $R_{EQ} = 2$ Ω.

Bo2's Drill Exercise 3.12

Find the Norton equivalent of the circuit below.

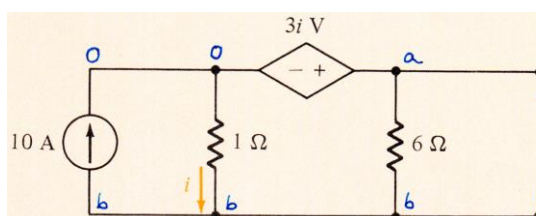


$\text{sth}("e,1,0,12:r6,1,2,6:j,0,2,3:r3,2,0,3",2,0)$

Choose DC. Via **ino** we find $I_{NO} = 8$ A. Via **req** we find $R_{EQ} = 1$ Ω.

Bo2's Drill Exercise 3.9

Find the Thévenin equivalent of the circuit below.

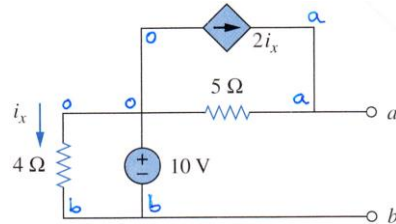


`s\th("j,b,0,10:r1,0,b,1:e,a,0,3:r1:r6,a,b,6",a,b)`

Choose DC. Via `vth` we find $V_{TH} = 24 \text{ V}$. Via `req` we find $R_{EQ} = 2.4 \Omega$.

AS2's Example 4.12

Find Norton equivalent of the circuit below.

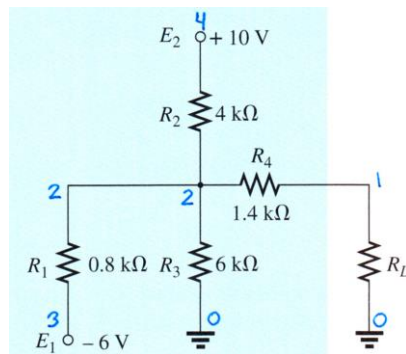


`s\th("rx,0,b,4:e,0,b,10:r5,0,a,5:j,0,a,2irx",a,b)`

Choose DC. Via `ino` we find $I_{NO} = 7 \text{ A}$. Via `req` we find $R_{EQ} = 5 \Omega$.

B11's Example 9.10 (Hidden source)

Find the Thévenin equivalent of the circuit below.

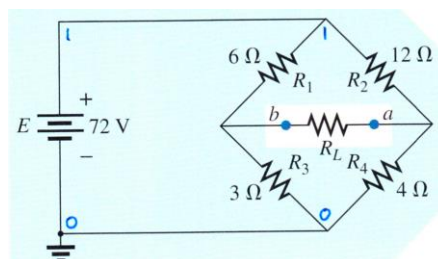


`true→s\si:s\th("e1,3,0,-6:e2,4,0,10:r1,2,3,0.8k:r2,2,4,4k:r3,2,0,6k:r4,2,1,1.4k",1,0)`

Choose DC. Via `vth` we find $V_{TH} = 3 \text{ V}$. Via `req` we find $R_{EQ} = 2 \text{ k}\Omega$.

B11's Example 9.9

Find the Thévenin equivalent of the circuit below, as seen from the R_L resistor.

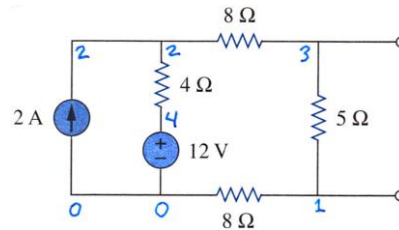


`s\th("e,1,0,72:r1,1,b,6:r2,1,a,12:r3,0,b,3:r4,0,a,4",b,a)`

Choose DC. Via `vth` we find $V_{TH} = 6 \text{ V}$. Via `req` we find $R_{EQ} = 5 \Omega$.

AS2's Example 4.11

Find the Norton equivalent of the circuit below.

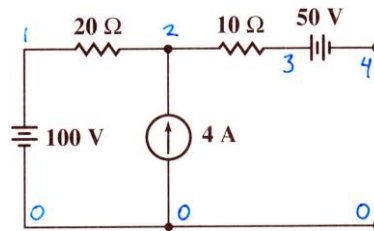


$s\backslash th("j,0,2,2:e,4,0,12:r1,2,4,4:r2,2,3,8:r3,0,1,8:r4,3,1,5",3,1)$

Choose DC. Via **ino** we find $I_{NO} = 1$ A. Via **req** we find $R_{EQ} = 4 \Omega$.

HK5's Drill Problem 2-8a

Find the Thévenin equivalent of the circuit below.

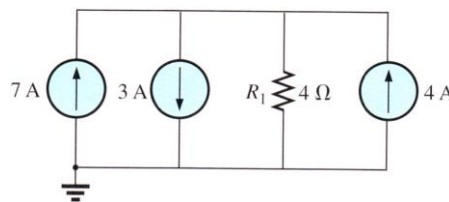


$s\backslash th("e1,1,0,100:r2,1,2,20:j,0,2,4:r1,2,3,10:e5,3,4,50",4,0)$

Choose DC. Via **vth** we find $V_{TH} = 130$ V. Via **req** we find $R_{EQ} = 30 \Omega$.

B11's Example 8.7

Find Norton equivalent of the circuit below.

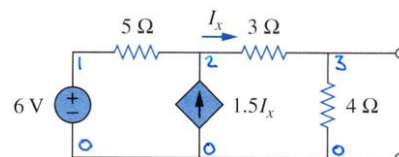


$s\backslash th("j7,0,1,7:j3,1,0,3:r1,1,0,4:j4,0,1,4",1,0)$

Choose DC. Via **ino** we find $I_{NO} = 8$ A. Via **req** we find $R_{EQ} = 4 \Omega$.

AS2's Practice Problem 4.9

Find the Thévenin equivalent of the circuit below.

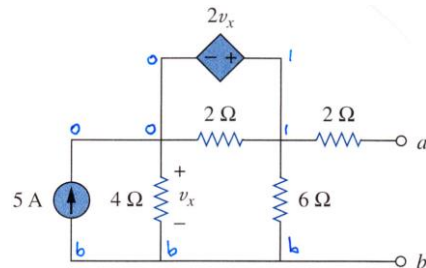


$s\backslash th("e,1,0,6:r5,1,2,5:rx,2,3,3:j,0,2,1.5irx:r4,3,0,4",3,0)$

Choose DC. Via v_{th} we find $V_{TH} = 5.33$ V. Via req we find $R_{EQ} = 0.44$ Ω .

AS2's Example 4.9

Find the Thévenin equivalent of the circuit below.

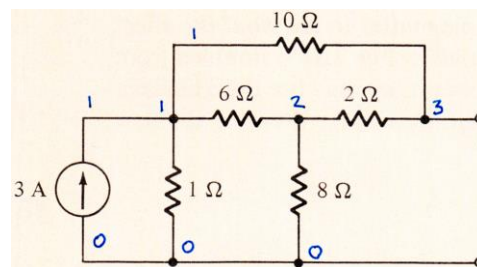


$s\backslash th("j,b,0,5:rx,0,b,4:r1,0,1,2:r2,1,b,6:r3,1,a,2:e,1,0,2vr_x",a,b)$

Choose DC. Via v_{th} we find $V_{TH} = 20$ V. Via req we find $R_{EQ} = 6$ Ω .

Bo2's Drill Exercise 3.8

Find the Thévenin equivalent of the circuit below.

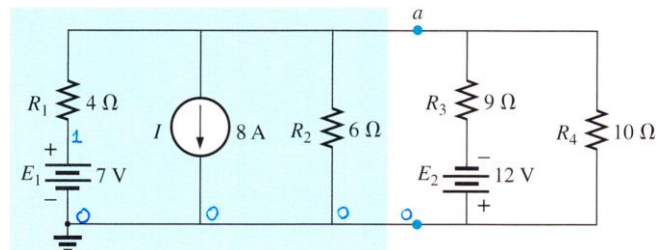


$s\backslash th("j,0,1,3:r1,0,1,1:r6,1,2,6:r10,1,3,10:r8,2,0,8:r2,2,3,2",3,0)$

Choose DC. Via v_{th} we find $V_{TH} = 2$ V. Via req we find $R_{EQ} = 4$ Ω .

B11's Example 9.13

Find Norton equivalent of the shaded part of the circuit.



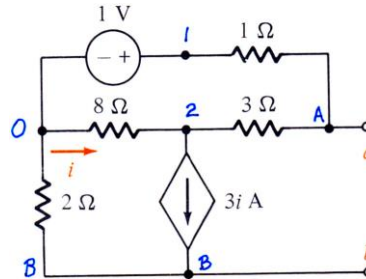
$s\backslash th("e1,1,0,7:r1,1,a,4:j,a,0,8:r2,a,0,6",a,0)$

Choose DC. Via ino we find $I_{NO} = -6.25$ A. Via req we find $R_{EQ} = 2.4$ Ω .

Some circuit theory books – and some professors – find it entertaining or instructive to surprise unsuspecting students with tricky problems, like the two we solve below.

Bo2's Example 3.11 (Tricky)

Find the Norton equivalent of the circuit below.



Since you do not know beforehand that this is a tricky problem, you go for the usual:

```
s\th("r2,0,b,2:r8,0,2,8:r3,2,a,3:r1,1,a,1:e,1,0,1:j,2,b,3ir8",a,b)
```

You choose DC, press Enter and wait. Symbolator reports the calculator was unable to solve the equations. This often means there is a division by zero somewhere.

Clean the variables from the MAIN folder and try again, this time – following Symbolator's advise –using a symbolic value. We chose to use **x** instead of **3** in the dependent source.

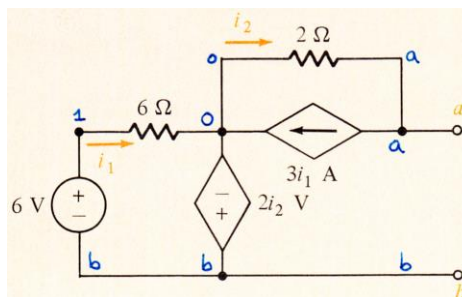
```
s\th("r2,0,b,2:r8,0,2,8:r3,2,a,3:r1,1,a,1:e,1,0,1:j,2,b,x*ir8",a,b)
```

Now the calculator can solve just fine. Evaluate the answers: the expression for **ino** is fine, but the expression for **req**, $(9*x-35)/(4*(x-3))$, will result in a division by zero.

Via **Define x=3: {ino,req}** we find that $I_{NO} = 1$ A, and R_{EQ} is undefined. This means the equivalent resistance is, for practical purposes, infinite. Your idea of fun, right?

Bo2's Drill Exercise 3.13 (Tricky)

Below is another tricky problem. Find the Norton equivalent of the circuit below.



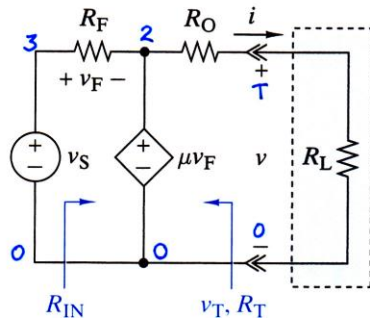
After trial and error, we use **x2** and **x3** instead of **2** and **3** in the dependent sources.

```
s\th("ei,1,b,6:ed,b,0,x2*ir2:r1,1,0,6:r2,0,a,2:jd,a,0,x3*ir1",a,b)
```

Exploring the answers, we see that the denominator of the expression for r_{eq} , $(x2*x3-6)$, is zero, which results in a division by zero. Via $2 \rightarrow x2: 3 \rightarrow x3: \{ino, req\}$ we find that $I_{NO} = -3A$, and R_{EQ} is undefined or, for practical purposes, infinite.

TR5's Exercise 4-6 (Symbolic)

Find the input resistance and output Thévenin equivalent circuit of the circuit



$$R_{IN} = (1 + \mu)R_F$$

$$v_T = \frac{\mu}{\mu + 1} v_s$$

$$R_T = R_O$$

First let's find the input resistance, R_{IN} , e.g. the resistance as seen by the v_s source. We want to use μ for the constant in the dependent source, so we set the `s\si` flag to false to avoid confusion with the SI unit for micro.

```
false→s\si:s\dc("ei,3,0,vs:rf,3,2,rf:ro,2,t,ro:ed,2,0,m*vrf"):rei
```

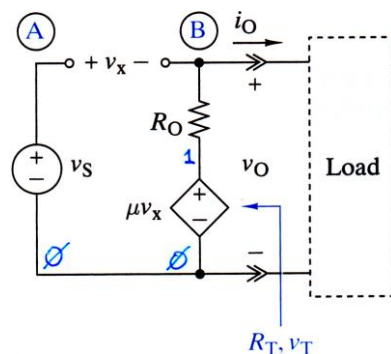
We get $rf*(\mu+1)$, which is correct. The textbook's answers are shown right of the circuit schematic. Now we find the output Thévenin equivalent circuit as seen by R_L .

```
s\th("ei,3,0,vs:rf,3,2,rf:ro,2,t,ro:ed,2,0,m*vrf",t,0):{vth,req}
```

We get $\{vs*\mu/(\mu+1),ro\}$, which is correct, as can be seen in the textbook's answers for v_T and R_T , shown right of the circuit schematic above.

TR5's Example 4-8 (Symbolic)

Find the Thévenin equivalent as seen by the load.



My answer starts by defining v_x as $va-vb$. This is done thus:

```
Define vx=va-vb
```

Now I run the `th` script, with the circuit description shown below:

```
false→s\si:s\th("ei,a,0,vs:ed,1,0,m*(vx):ro,b,1,ro",b,0):{vth,req}
```

The answers we get, $\{v_s * \mu / (\mu + 1), r_o / (\mu + 1)\}$, are correct, as can be seen by comparing them to those in the book:

$$v_T = \frac{\mu v_s}{\mu + 1} \quad \text{and} \quad R_T = \frac{R_O}{\mu + 1}$$

I am not sure there is any other circuit simulator for calculators that can do this.

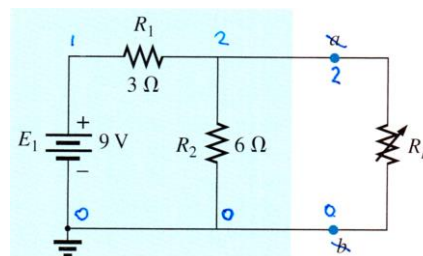
RL Problems

th saves the current, voltage drop and power in a load as a function of load value L in variables irL, vrL and prL, respectively.

One type of problems that books and professors like to present to students when teaching the Thévenin / Norton equivalents is what I like to call RL problems. A typical RL problem statement goes like this: "First, reduce the circuit, as seen by resistor RL, to its Thévenin or Norton equivalent. Then, find the value of the voltage drop, current and/or power consumed in the load resistor RL if its value is (whatever) ohms." Since this is such a typical problem, Symbolator helps you solve them. Right after a Thévenin or Norton equivalent is found, Symbolator will store a series of variables related to a RL connected among the equivalent's terminals: its current is stored in **irL**, its voltage drop in **vrL**, and its consumed power in **prL**, all in function of the value of the load, **L**. These are good as long as the load is the only thing connected to the equivalent.

B11's Example 9.6

Find the Thévenin equivalent circuit for the network in the shaded area. Then find the current through RL for RL values of 2Ω, 10Ω and 100Ω.



My answer below. First we find the Thévenin equivalent.

```
s\th("e1,1,0,9:r1,1,2,3:r2,2,0,6",2,0):{vth,req}
```

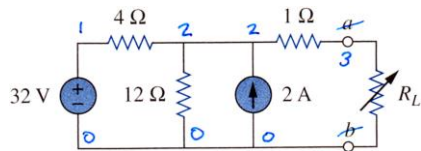
The answer, $\{6, 2\}$, is correct. Now we find the values of irL for the different values.

```
{irL|L=2.,irL|L=10.,irL|L=100.}
```

Where | is the "given" operator. The answers, $\{1.5, .5, .059\}$, are correct.

AS2's Example 4.8

Find the Thévenin equivalent of the circuit shown to the left of terminals a-b. Then find the current through RL = 6, 16 and 36Ω.



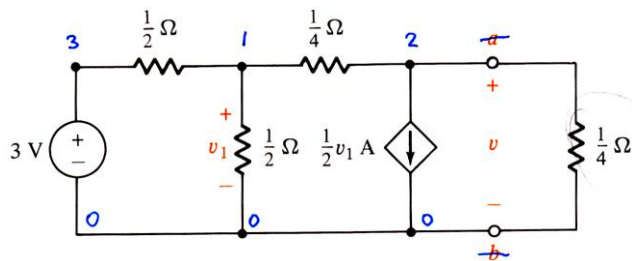
My solution below.

```
s\th("e,1,0,32.:r4,1,2,4:r12,2,0,12:j,0,2,2:r1,2,3,1",3,0):
{vth,req,irL|L=6,irL|L=16,irL|L=36}
```

The answer, **{30,4,3,1.5,.75}**, is correct.

Bo2's Example 3.10

Find the Norton equivalent of the circuit left of the a-b terminals, and then find the voltage drop and the current through the $\frac{1}{4}\Omega$ resistor. My one-line solution below.

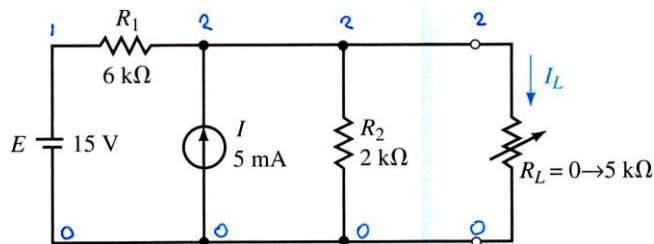


```
s\th("e,3,0,3:r31,3,1,1/2:r10,1,0,1/2:r12,1,2,1/4:j,2,0,v1/2",2,0):
{ino,req,vrL|L=1/4,irL|L=1/4}
```

The book gives the answers as fractions. We get it right: **{21/8,4/9,21/50,42/25}**.

RM3's Example 9-7

Find the Norton equivalent of the circuit external to R_L . Then determine the load current I_L when $R_L = 0\Omega$, $2k\Omega$ and $5k\Omega$. My one-line solution below.

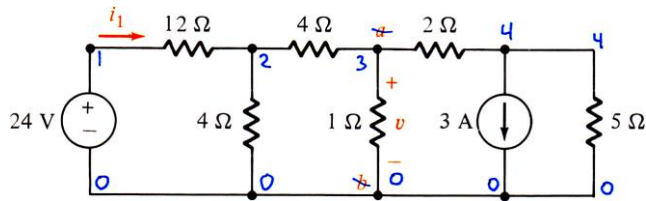


```
true→s\si:s\th("e,1,0,15.:r1,1,2,6k:j,0,2,5m:r2,2,0,2k",2,0):
{ino,req, irL|L=0,irL|L=2000,irL|L=5000}
```

The answer, **{.0075,1500,.,0075,.00321,.00173}**, is correct.

Bo2's Example 3.5

Find the Norton equivalent of the circuit external to the 1Ω resistor. Then determine the voltage drop across this 1Ω resistor.



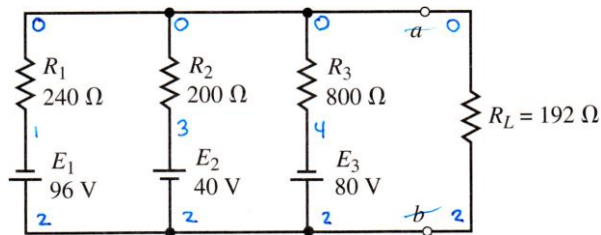
My solution below.

```
s\th("e,1,0,24:r12,1,2,12:r20,2,0,4:r32,2,3,4:r34,3,4,2:
j,4,0,3:r40,4,0,5",3,0):{ino,req,vrL|L=1}
```

The answer, **$\{-9/7, 7/2, -1\}$** , is correct.

RM3's Example 9-13

Use Millman's Theorem to simplify the circuit left of a-b so that there is only one voltage and one resistor. Then find the current in the load resistor R_L .



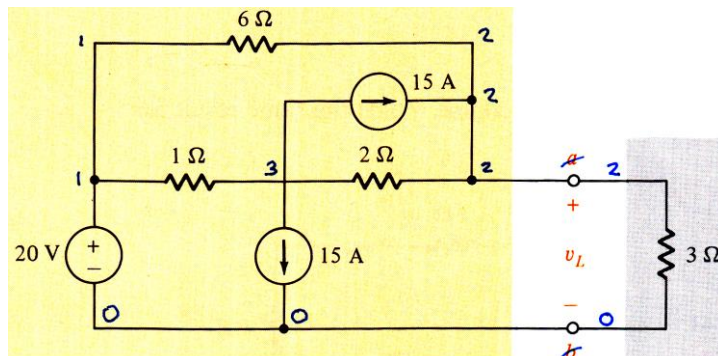
I don't know Millman's Theorem, but in my book this is called the Thévenin equivalent.

```
s\th("r1,0,1,240.:e1,2,1,96:r2,0,3,200:e2,3,2,40:
r3,0,4,800:e3,2,4,80",2,0):{vth,req,irL|L=192}
```

The answer, **$\{28.8, 96, .1\}$** , is correct.

Bo2's Example 3.7

Find the Thévenin equivalent for the circuit left of a-b. Then find the voltage across the 3Ω resistor. My answer is presented below.



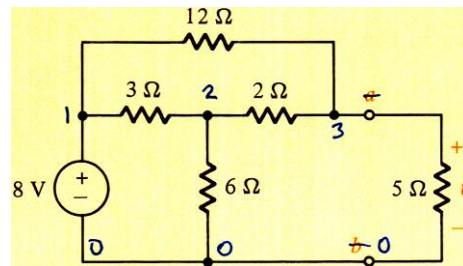
```
s\th("e,1,0,20:r6,1,2,6:r1,1,3,1:r2,3,2,2:j32,3,2,15:
j30,3,0,15",2,0):{vth,req,vrL|L=3,vrL|L=6.}
```

The answer we find, **$\{30, 2, 18, 22.5\}$** , is correct.

Bo2's Drill Exercise 3.7

Find the Thévenin equivalent for the circuit left of a-b. Then find the voltage v .

My answer below.

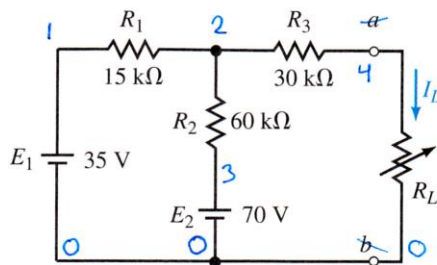


```
s\th("e,1,0,8:r3,1,2,3:r12,1,3,12:r6,2,0,6:r2,2,3,2",3,0):{vth,req,vrL|L=5.}
```

The answer, **{6,3,3.75}**, is correct.

RM3's Practice Problem 9.5

Find the Norton equivalent external to R_L in the circuit. Solve for the current I_L when $R_L = 0, 10 \text{ k}\Omega, 50 \text{ k}\Omega$, and $100 \text{ k}\Omega$.



Answers

$R_N = 42 \text{ k}\Omega, I_N = 1.00 \text{ mA}$

For $R_L = 0$: $I_L = 1.00 \text{ mA}$

For $R_L = 10 \text{ k}\Omega$: $I_L = 0.808 \text{ mA}$

For $R_L = 50 \text{ k}\Omega$: $I_L = 0.457 \text{ mA}$

For $R_L = 100 \text{ k}\Omega$: $I_L = 0.296 \text{ mA}$

My solution below.

```
true→s\si:s\th("e1,1,0,35:r1,1,2,15k:r2,2,3,60k:e2,3,0,70:r3,2,4,30k",4,0):{ino,req,irL|L=0,irL|L=1E4,irL|L=5E4,irL|L=1E5}
```

The answer, **{.001,42000,.001,.000808,.000457,.00296}**, is correct.

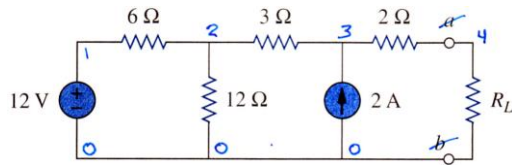
Power transfer problems

Another type of problem that is often associated to the Thévenin / Norton equivalents is that regarding the power transfer to a load, particularly the maximum power transfer possible. Maximum power is transferred when the load R_L is equal to the R_{EQ} of the circuit's equivalent. Symbolator stores, in variable **pMax**, the maximum power that can be delivered by a Thévenin / Norton equivalent circuit; and also stores in **prL** the power transferred to the load as a function of its value, **L**. Let's see some examples.

th saves the
maximum
power in **pMax**
and the power
as a function of
load **L** in **prL**

AS2's Example 4.13

Find the R_L value for maximum power transfer and the maximum power transferred.



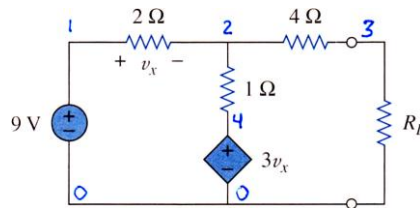
R_L for maximum transfer is **rEq**. The maximum power is in **pMax**. My solution below:

```
s\th("e,1,0,12:r6,1,2,6:r12,2,0,12:r3,2,3,3:
j,0,3,2:r2,3,4,2",4,0):approx({req,pmax})
```

The answer, **{9.,13.44}**, is correct: the maximum transfer of power occurs when the load is 9Ω . At this point, the power transferred is 13.44W. Now let's solve another one.

AS2's Practice Problem 4.13

Find the R_L value for maximum power transfer and the maximum power transferred.

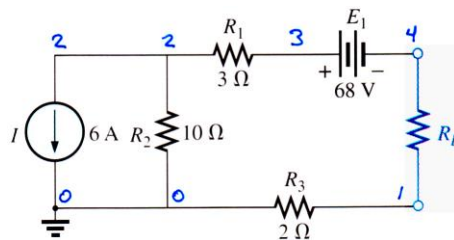


```
s\th("ei,1,0,9:rx,1,2,2:r1,2,4,1:ed,4,0,3vr:4,2,3,4",3,0):approx({req,pmax})
```

The logic of this problem is identical to the previous one. The answer is **{4.22,2.901}**.

B11's Example 9.17

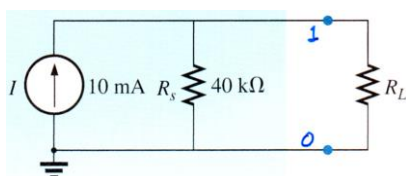
Find the R_L value for maximum power transfer and the maximum power transferred.



```
s\th("j,2,0,6:r2,2,0,10:r1,2,3,3:r3,0,1,2:e,3,4,68",4,1):approx({req,pmax})
```

Same thing. Answer is **{15.,273.07}**. Let's now see one that is a little different.

B11's Example 9.15



- Find the load resistance that will result in maximum power transfer to the load, and find the maximum power delivered.
- If the load were changed to $68\text{ k}\Omega$, would you expect a fairly high level of power transfer to the load based on the results of part (a)? What would the new power level be? Is your initial assumption verified?
- If the load were changed to $8.2\text{ k}\Omega$, would you expect a fairly high level of power transfer to the load based on the results of part (a)? What would the new power level be? Is your initial assumption verified?

My one-line solution to all three questions is presented below.

```
true→s\si:s\th("j,0,1,10m:r,1,0,40k",1,0):
{req,pmax,prl|I=68000.,prl|I=8200.}
```

The answer we obtain, **{40000,1,.93,.57}**, is correct. Let's deconstruct it.

Part (a) is answered by the first two values: a 40K Ω resistor as load would receive 1W power. Since this is the maximum – this is the most that any load could receive ever.

Parts (b) is answered by the third value. Making use of the variable **prL**, which contains the power delivered by the circuit equivalent to the load, as a function of the load value **L**, we find that a load of 68k Ω receives .93W, which is less than the maximum.

Parts (c) is answered in similar manner by the fourth value. A load of 8.2k Ω receives .57W, which is less than the maximum. Any resistance other than 40K gets less power.

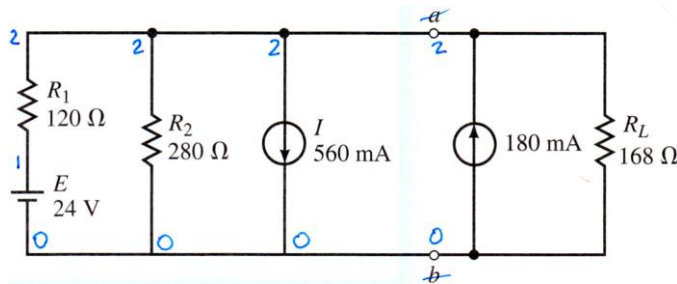
Automatic circuit equivalent

At risk of promoting vagrancy among EE students, let me show you one more goodie of the **th** script: once it has found the Norton equivalent of a circuit, it will automatically write for you the description of the equivalent circuit and store it in a variable, **eqcir**, to save you time just in case you need to run any simulation using it. The following example illustrates the use of this laziness-promoting feature of **th**.

th saves a description of the equivalent circuit in a variable called **eqcir**

RM3's Example 9-8

Find the Norton equivalent of the circuit left of a-b; then find the current through R_L .



Let's first find the circuit equivalent:

```
true→s\si:s\th("e,1,0,24:r1,1,2,120:r2,2,0,280:j,2,0,560m",2,0):{ino,req}
```

The answer, **{-.36,84.}**, is correct. Now to the second part of the question. In order to find the current through R_L , we cannot use – as we did before – the variable **irL**, because now the load is not the only thing connected to the terminals of the circuit equivalent: there is also a current source. The expressions stored in **irL**, **vrL** and **prL**, are only valid when a load R_L is the only thing connected to the circuit equivalent.

So we have no other option than to run a simulation. The fastest way to do this is to use the contents of the variable **eqcir**. Inside this variable, we find the following circuit description: "jN,0,n,iNo:rE,n,0,rEq:rL,n,0,L". The values for iNo and rEq are



already in the calculator's memory. So we change the value of the load to 168Ω , and add the 180mA source flowing from node **0** to node **n**, the nodes used by **eqcir**.

```
s\dc("jN,0,n,iNo:rE,n,0,rEq:rL,n,0,168:j,0,n,180m"):irL
```

The answer, **-06**, is correct: there is a current of 60mA flowing through R_L from 0 to n. Using **eqcir** is meant to save you time. If you find it confusing to use, just don't use it.

Element: o

Ideal Op Amp

Describing an ideal operational amplifier

There are many types of operational amplifiers. Symbolator can simulate the ideal, linear type. In Symbolator, an ideal Op Amp is described as follows: first the name of the Op Amp, which must start with the letter **o**, and a coma; second the name of an input node, another coma; third the name of the other input node, another coma; and fourth the name of the output node. The polarity of the input nodes does not matter.

*This is how
ideal Op Amps
are described in
Symbolator.*

An ideal op amp called **o1**, whose input nodes are **2** and **3**, and whose output node is node **5**, would be described thus: **o1,2,3,5**. And an ideal op amp called **o**, whose input nodes are **p** and **n**, and whose output node is also called **o**, would be: **o,p,n,o**

Simulation answers

For each ideal op amp in a circuit, Symbolator will store the following answers:

- The *voltage of each of its nodes* with reference to the ground is stored in a variable called **v** and the name of the node. For example, for node **o**, the voltage is in **vo**.
- The *current through the output node*, flowing from the output node outwards. For example, the current leaving the output node of an op amp called **o** is stored in **io**.
- The *power consumed* by the (visible part of) the op amp is stored in a variable called **p** and the name of the op amp. For example, for an op amp called **o2**, the power consumed is **po2**; the power *delivered* is the negative of that, **-po2**.

A node by any other name

Since I was born to be bad, I like to play with how I name nodes in order to get the answers as close to the book as possible. So, do not be surprised if I call one op amp **o** and then also name its output node **o**. That way I can will ask for **vo** and **io**, and get the output voltage and current⁶. But realize this: in asking for **vo**, we are asking for the voltage in node **o**. And in asking for **io**, the **o** stands for the element **o**.

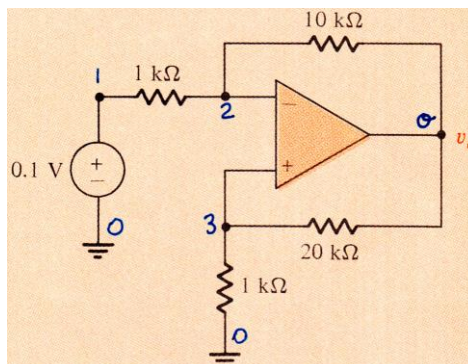
When deciding the name of nodes, however, you should be aware of the fact that, for each node **#**, Symbolator will create a variable called **v#**, to store the voltage of that node. If you are not careful, you may inadvertently create a problem for Symbolator. Particularly, you should avoid describing a source thus: **e,#,0,v#**, where **#** is anything. For example, if you describe a source as **e1,1,0,v1** Symbolator will define the voltage of node 1, namely **v1** as having the value you provided: **v1**. The resulting equation, **v1=v1** is discarded by the calculator, which leaves Symbolator one equation short.

⁶ Read this wearing amber sunglasses and playing in your mind the beginning of the *CSI: Miami* theme.

Solved problems

Bo2's Drill Exercise 3.2

For the op-amp circuit shown, find v_o and the power absorbed by the op amp.

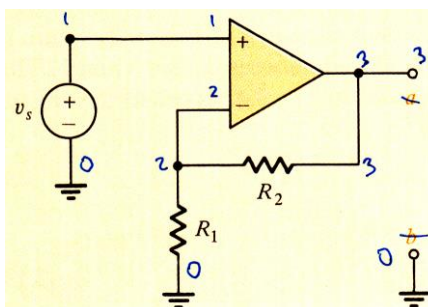


`true→s\si:s\dc("e,1,0,.1:r12,1,2,1k:r2o,2,o,10k:r3o,3,0,1k:r3o,3,o,20k:o,3,2,o"):{vo,po}`

The answer, **$\{-2.1, -.00063\}$** , is correct. The voltage in node o is -2.1V, and the op amp is absorbing -.63mW, which is to say it is delivering .63mW of power to the circuit.

Bo2's Drill Exercise 3.11 (Thévenin)

Find the Thévenin equivalent.



`s\th("e,1,0,vs:r1,2,0,r1:r2,2,3,r2:o,1,2,3",3,0)`

The **th** script tells us it found the Thévenin voltage, but could not find the Norton current. This is not a surprise, since an ideal op amp has zero output resistance and a fixed voltage, an infinite current when short-circuited. So the Thévenin equivalent is given by V_{TH} and no resistance (or $R_{EQ} = 0\Omega$). Evaluating **vth** results in

$$\frac{(r1 + r2) vs}{r1}$$

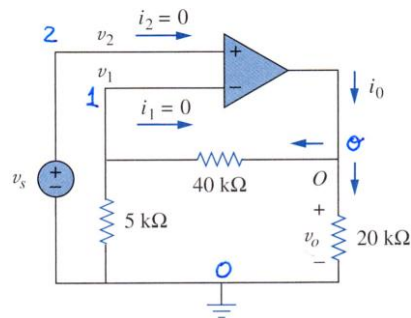
This is correct, as can be seen by comparing it to the book's answer, shown below.

$$\left(1 + \frac{R_2}{R_1}\right) v_s$$

The Thevenin resistance, as explained above, is 0Ω .

AS2's Example 5.2

Find v_o and i_o .

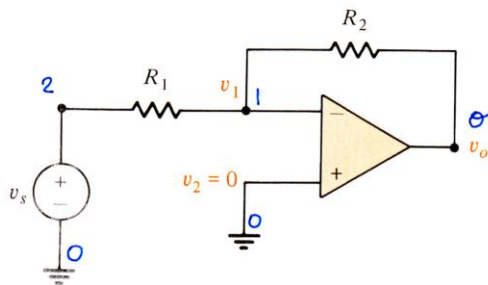


`true⇒s\si:s\dc("e,2,0,1:r5,1,0,5k:r4,1,o,40k:r2,o,0,20k:o,2,1,o"): {vo,io}`

The answer, **{9,.,00065}**, is correct: v_o is 9V and i_o is 0.65mA

Bo2's Figure 3.3 (Inverting)

Find the gain of the overall circuit, v_o/v_s .



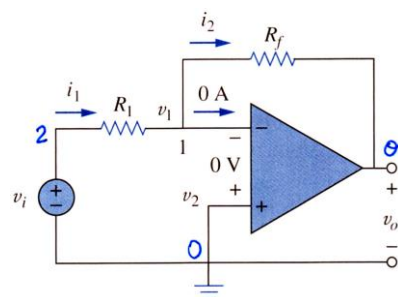
$$\frac{v_o}{v_s} = -\frac{R_2}{R_1}$$

`s\dc("e,2,0,vs:r1,2,1,r1:r2,1,o,r2:o,1,0,o"):vo/vs`

We get **-r2/r1**, which is correct, as can be seen in the book's answer above.

AS2's Figure 5.10 (Inverting)

Find v_o .



$$v_o = -\frac{R_f}{R_1} v_i$$

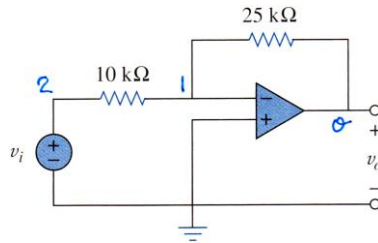
`s\dc("e,2,0,vi:r1,2,1,r1:rf,1,o,rf:o,0,1,o"):vo`

This problem is almost identical to the one above. The answer we get is correct:

$$-\frac{rf}{r1} vi$$

AS2's Example 5.3 (Inverting)

If v_i is 0.5V, calculate the output voltage v_o and the current in the 10k Ω resistor.

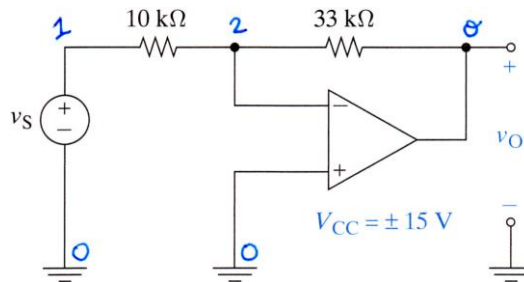


```
true→s\si:s\dc("e,2,0,.5:r1,2,1,10k:rf,1,o,25k:o,0,1,o"):{vo,ir1}
```

The answer, **$\{-1.25, 5.E-5\}$** , is correct.

TR5's Exercise 4-11 (Inverting)

Find v_o when v_s is 2V, -4V and 6V. Notice the output of the op amp is limited to ± 15 V.



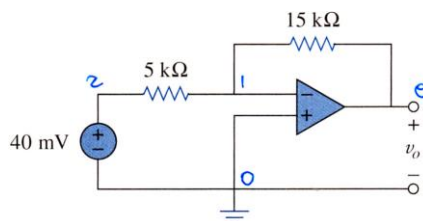
This may be the only non-linear problem you will see in this book, because I solved it before I realized it included the ± 15 V constraint. But anyway, here it goes.

```
true→s\si:s\dc("e,1,0,vs:r1,1,2,10k:r2,2,o,33k:o,0,2,o") {vo|vs=2.,vo|vs=-4.,vo|vs=6.}
```

The answer, **$\{-6.6, 13.2, -19.8\}$** , is correct within the linear realm, but since the output is constrained to no more than 15V or less than -15V, the answer is $v_o = -15$ V for $v_s = 6$ V.

AS2's Practice Problem 5.3 (Inverting)

Find the output voltage of the op amp (e.g. v_o) and calculate the current through the feedback resistor (e.g. the 15k Ω resistor).



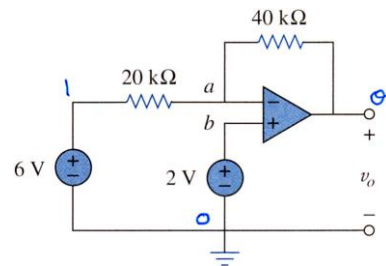
My answer below. Notice we used the m for milli in the value of the voltage source.

```
true→s\si:s\dc("e,2,0,40m:r1,2,1,5k:rf,1,o,15k:o,0,1,o"):approx({vo,irf})
```

The answer, **$\{-12, 8.E-6\}$** , is correct.

AS2's Example 5.4 (Inverting)

Determine v_o .

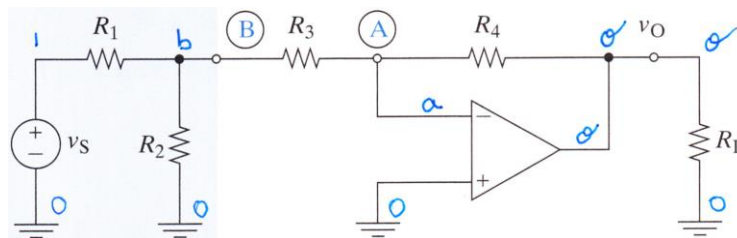


`true→s\si:s\dc("e6,1,0,6:r1,1,a,20k:e2,b,0,2:rf,a,o,40k:o,a,b,o"):vo`

The answer, **-6**, is correct.

TR5's Example 4-14 (Inverting)

Find the input-output relationship of the circuit below.



The 'formal' way to *symbolate* this circuit would be as described below.

`s\dc("e,1,0,vs:r1,1,b,r1:r2,b,0,r2:r3,b,a,r3:r4,a,o,r4:o,0,a,o:rl,o,0,rl")`

However, since we are only interested in the ratio of the input to the output, a smarter (and faster – as in 49 seconds instead of 55 seconds) way to *symbolate* it is this:

`s\dc("e,1,0,1:r1,1,b,r1:r2,b,0,r2:r3,b,a,r3:r4,a,o,r4:o,0,a,o:rl,o,0,1")`

When we evaluate v_o/v_1 , both approaches get the same answer, shown left below:

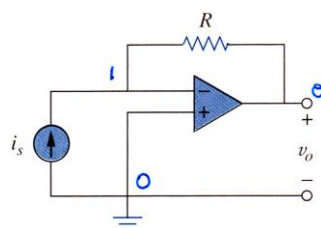
$$\frac{-r_2 r_4}{r_1 (r_2 + r_3) + r_2 r_3}$$

$$-\frac{R_2 R_4}{R_1 R_2 + R_1 R_3 + R_2 R_3}$$

which is correct, as can be seen by comparing it to the book's answer, shown right.

AS2's Practice Problem 5.4a (Transresistance)

This is a current-to-voltage converter, also called a *transresistance amplifier*. Find v_o/i_s .

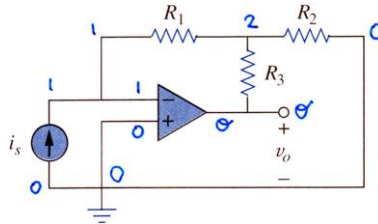


`s\dc("j,0,1,is:r,1,o,r:o,0,1,o"):vo/is`

The answer, $-r$, is correct.

AS2's Practice Problem 5.4b (Transresistance)

This is another *transresistance amplifier*. Again, find v_o/i_s .



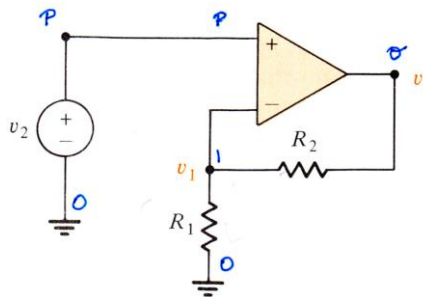
`s\dc("j,0,1,is:r1,1,2,r1:r2,2,0,r2:r3,2,o,r3:o,0,1,o")`

Asking `expand(vo/is)` get $-r1*r3/r2-r1-r3$, which is equivalent to the book's answer.

$$\frac{v_o}{i_s} = -R_1 \left(1 + \frac{R_3}{R_1} + \frac{R_3}{R_2} \right)$$

Bo2's Example 3.1 (Non-Inverting Amplifier)

Find v_o/v_1 .

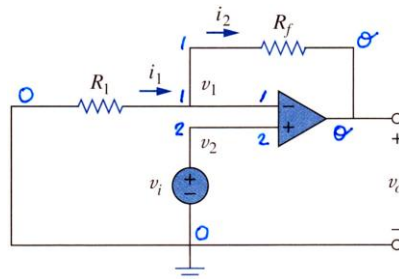


`s\dc("e,p,0,v2:r1,1,0,r1:r2,1,o,r2:o,p,1,o"):vo/v1`

We get $1+r2/r1$, which is correct.

AS2's Figure 5.16 (Non-Inverting Amplifier)

Find v_o .



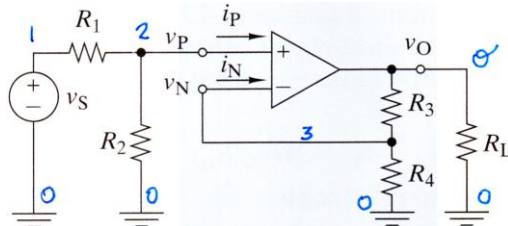
$$v_o = \left(1 + \frac{R_f}{R_1} \right) v_i$$

`s\dc("e,2,0,vi:r1,0,1,r1:rf,1,o,rf:o,2,1,o"):vo`

We get the right answer (below), an expression equivalent to the book's answer.

$$\frac{(r1 + rf)}{r1} v_i$$

TR5's Example 4-13 (Non-Inverting Amplifier)



We can solve this problem in one simulation, as shown below.

```
s\dc("e,1,0,vs:r1,1,2,r1:r2,2,0,r2:o,2,3,o:r3,o,3,r3:r4,3,0,r4"):vo/vs
```

$$\frac{r2 (r3 + r4)}{(r1 + r2) r4}$$

which is correct, as can be seen by comparing it to the book's answer, shown below:

$$\left[\frac{R_2}{R_1 + R_2} \right] \left[\frac{R_3 + R_4}{R_4} \right]$$

We can also solve it in stages, as shown below. First, simulate the left half.

```
s\dc("e,1,0,vs:r1,1,2,r1:r2,2,0,r2"):v2/vs
```

$$\frac{r2}{(r1 + r2)}$$

Which is correct for this part. Then, simulate the right half.

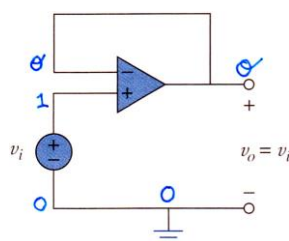
```
s\dc("e,2,0,1:o,2,3,o:r3,o,3,r3:r4,3,0,r4"):vo/v2
```

$$\frac{(r3 + r4)}{r4}$$

Which is correct for this part. The product of these two partial answers produces the same expression shown above after the big simulation, and is the right answer.

AS2's Figure 5.17 (Voltage Follower)

Find v_o .



s\dc("e,1,0,vi:o,1,o,o"):vo

The answer, **vi**, is correct.

TR5's Figure 4-32 (Voltage Follower)

The circuits in Figure 4-32 have $v_S = 1.5$ V, $R_S = 2$ k Ω , and $R_L = 1$ k Ω . Compute the maximum power available from the source. Compute the power absorbed by the load resistor in the direct connection in Figure 4-32(b) and in the voltage follower circuit in Figure 4-32(a). Discuss any differences.

Answers: 281 μ W; 250 μ W; 2250 μ W

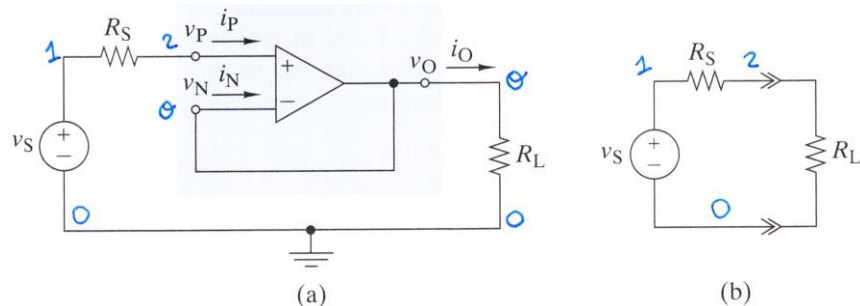


FIGURE 4-32 (a) S load interface with a voltage follower. (b) Interface without the voltage follower.

Alright, so first we simulate the (b) circuit, to find its maximum power and load power:

true→s\si:s\th("e,1,0,1.5:rs,1,2,2k",2,0):{pmax,prL|L=1000}

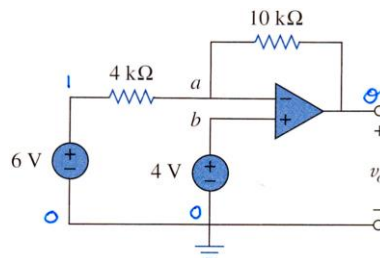
The answers we get, **{2.8125E-4,2.5E-4}**, are correct. Now we simulate the (a) circuit.

s\dc("e,1,0,1.5:rs,1,2,2k:o,2,o,o:rl,o,0,1k"):prL

The answer, **.00225**, is correct. The explanation to the apparent paradox that the load in (a) is drawing more power than the source in (b) seems able to provide evaporates once we remember that the ideal op amp shown in the schematic is only part of the truth: the real op amp has its own source of power, which provides the difference.

AS2's Example 5.5 (Inverting)

Find v_o .

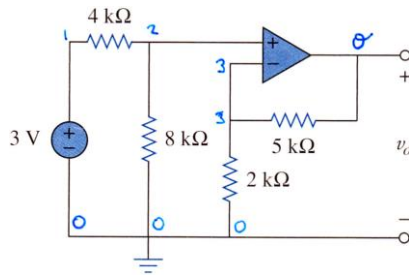


true→s\si:s\dc("e,1,0,6:r4,1,a,4k:r10,a,o,10k:e4,b,0,4:o,b,a,o"):vo

The answer, **-1**, is correct.

AS2's Practice Problem 5.5 (Non-Inverting)

Calculate v_o .

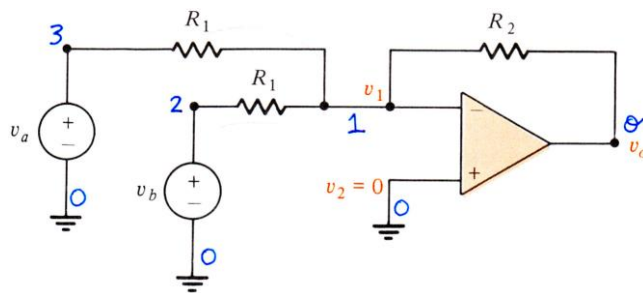


`true→s\si:s\dc("e,1,0,3:r4,1,2,4k:r8,2,0,8k:r2,3,0,2k:r5,3,0,5k:o,2,3,o")`

The answer for v_o , **7**, is correct.

Bo2's Example 3.2 (Adder or Summing)

Find v_o .



`s\dc("ea,3,0,va:eb,2,0,vb:r31,3,1,r1:r21,2,1,r1:r1o,1,0,r2:o,0,0,1,o"):vo`

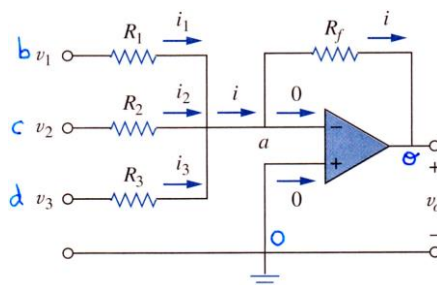
$$-\frac{r_2}{r_1}(v_a + v_b)$$

which is correct, as can be seen by comparing it to the book's answer, shown below:

$$v_o = -\frac{R_2}{R_1}(v_a + v_b)$$

AS2's Figure 5.21 (Adder or Summing)

Find v_o .



`s\dc("e1,b,0,v1:e2,c,0,v2:e3,d,0,v3:r1,b,a,r1:r2,c,a,r2:r3,d,a,r3:rf,a,o,rf:o,0,a,o"):expand(vo)`

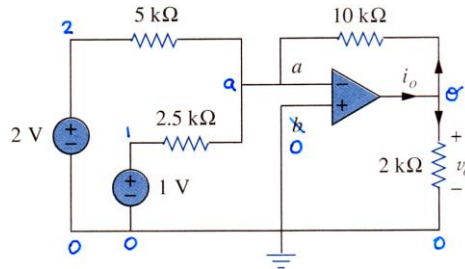
$$-r_f \frac{v_1}{r_1} - r_f \frac{v_2}{r_2} - r_f \frac{v_3}{r_3}$$

which is correct, as can be seen by comparing it to the book's answer, shown below:

$$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$$

AS2's Example 5.6 (Adder or Summing)

Find v_o and i_o .

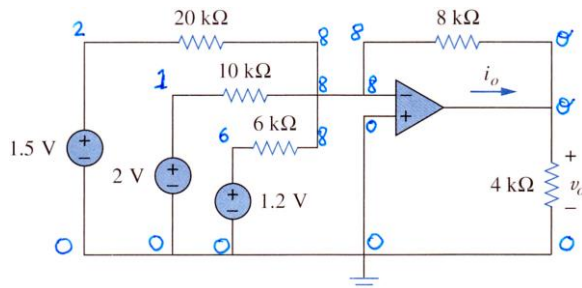


```
true→s\si:s\dc("e1,1,0,1:e2,2,0,2:r1,2,a,5k:
r2,1,a,2.5k:r3,a,o,10k:r4,o,0,2k:o,0,a,o"):{vo,io}
```

The answer, **$\{-8.,-.0048\}$** , is correct. Notice a current of 4.8mA is going into the op amp.

AS2's Practice Problem 5.6 (Adder or Summing)

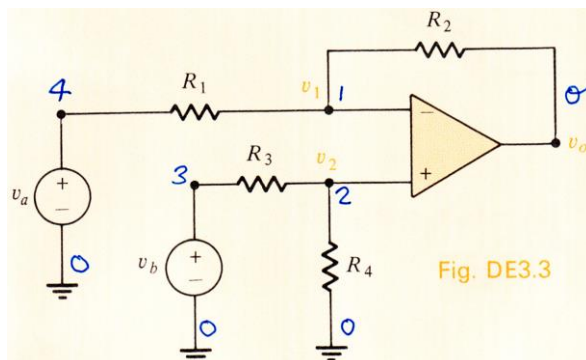
Find v_o and i_o .



```
true→s\si:s\dc("e2,2,0,1.5:e1,1,0,2:e6,6,0,1.2:r2,2,8,20k:
r1,1,8,10k:r6,6,8,6k:r8,8,o,8k:r4,o,0,4k:o,0,8,o"):{vo,io}
```

The answer, **$\{-3.8,-.001425\}$** , is correct. Again, the current is going into the op amp.

Bo2's Drill Exercise 3.3 (Difference or Differential)



Find v_o for the **difference-amplifier** circuit shown in Fig. DE3.3 when $R_3 = R_1$ and $R_4 = R_2$.

Answer: $\frac{R_2}{R_1} (v_b - v_a)$

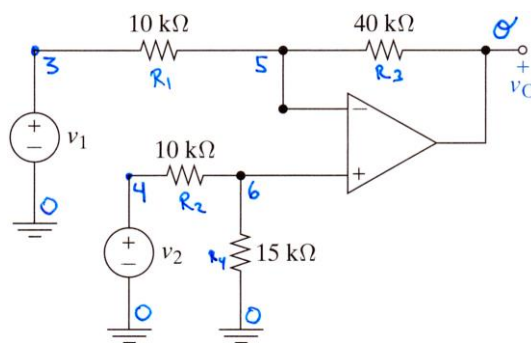
`s\dc("ea,4,0,va:eb,3,0,vb:r1,4,1,r1:r2,1,o,r2:r3,3,2,r1:r4,2,0,r2:o,2,1,o")`

Evaluating v_o we get the correct answer, equivalent to the book's answer above.

$$\frac{(v_b - v_a) r_2}{r_1}$$

TR5's Exercise 4-13 (Difference or Differential)

Find v_o .

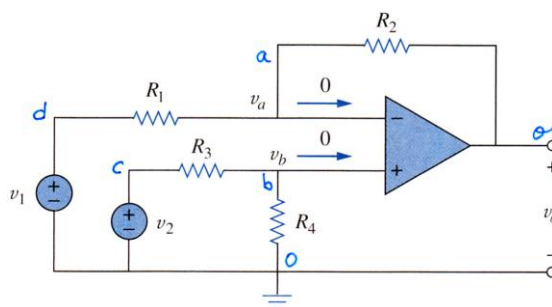


`s\dc("e1,3,0,v1:e2,4,0,v2:r1,3,5,10k:r2,4,6,10k:r3,5,o,40k:r4,6,0,15k:o,6,5,o")`

Evaluating v_o we get $3 v_2 - 4 v_1$, which is the correct answer.

AS2's Figure 5.24 (Difference or Differential)

Find v_o . (And keep it in the memory, for you will use it in the next three problems.)



`s\dc("e1,d,0,v1:e2,c,0,v2:r1,d,a,r1:r3,c,b,r3:r2,a,o,r2:r4,b,0,r4:o,b,a,o"):v_o`

$$\frac{r_1 r_4 v_2 - r_2 (r_3 v_1 + r_4 (v_1 - v_2))}{r_1 * (r_3 + r_4)}$$

This expression is equivalent to the book's answer, better formatted, shown below:

$$v_o = \frac{R_2}{R_1} \frac{(1 + R_1/R_2)}{(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$$

AS2's Figure 5.24 (Subtractor)

For the same circuit of the previous problem, find v_o when $R_1=R_2$ and $R_3=R_4$.

Since we already have the expression for v_o stored in the memory, we only do this:

```
expand(vo)|r2=r1 and r3=r4
```

The answer we get, **v_2-v_1** , is correct.

AS2's Example 5.7 (Difference or Differential)

Design an op amp circuit with inputs v_1 and v_2 such that $v_o = -5v_1 + 3v_2$.

My solution follows. The problem statement is a fancy way to say: for the same circuit of the previous problem, find what values of resistors you need to use if you want to get an output $v_o = -5v_1 + 3v_2$. Although this is not strictly a Symbolator problem, I present it here because it illustrates how Symbolator fits in such design problems.

First we take that part of v_o that is a factor of v_1 , and make it equal to -5. Thus:

```
Define v1=1:Define v2=0:expand(vo)=-5
```

We get **$-r_2/r_1=-5$** . Now make that part of v_o that is a factor of v_2 equal to 3. Thus:

```
Define v1=0:Define v2=1:expand(vo)=3
```

We get **$r_2*r_4/(r_1*(r_3+r_4))+r_4/(r_3+r_4)=3$** Now, since you have two equations, you can solve for two unknowns. Out of the four resistors whose values you can decide upon, two can be whatever you want. The book recommends you use $R_1 = 10K$ and $R_3=20K$. Now let's find the values for the other two resistors, R_2 and R_4 .

```
solve(ans(1) and ans(2),{r2,r4})|r1=10000 and r3=20000
```

The expression above assumes that `ans(1)` and `ans(2)` are pointing to the two equations we found before.⁷ We get **$r_2=50000$ and $r_4=20000$** . This is correct.

Now, if this problem was part of a test, I'd like to verify that the answer is correct. To confirm this, simulate the circuit using the four values given above for the resistors.

```
true→s\si:s\dc("e1,d,0,v1:e2,c,0,v2:r1,d,a,10k:r3,c,b,20k:r2,a,o,50k:r4,b,0,20k:o,b,a,o")
```

Evaluating `vo` gives us the desired output, **$3*v_2-5*v_1$** . The resistor values are correct.

AS2's Practice Problem 5.7 (Difference or Differential)

Design a difference amplifier with gain 4.

My solution follows. The problem statement, again, is just a fancy way to say: for the same circuit of the previous problem, find what values of resistors you need to use if you want to get an output $v_o = 4(v_2-v_1)$, or in other terms, $-4v_1 + 4v_2$. Same as before.

⁷ If you have no idea what I'm talking about, you should learn about `ans(#)` in the user's manual.

Define v1=1:Define v2=0:vo=-4

Define v1=0:Define v2=1:expand(vo)=4

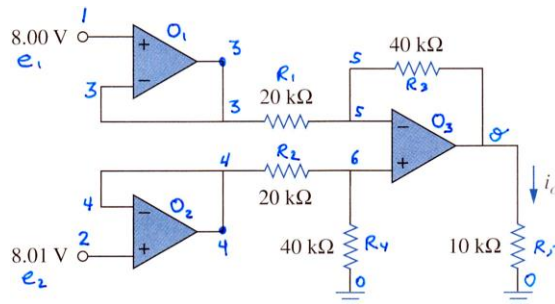
This time the book asks that you use $R_1 = 10\text{K}$ and $R_3 = 10\text{K}$. So we do that.

`solve(ans(1) and ans(2),{r2,r4})|r1=10000 and r3=10000`

We get **$r_2=40000$ and $r_4=40000$** , the correct values for the remaining resistors.

AS2's Practice Problem 5.8 (Instrumentation)

Find i_o .

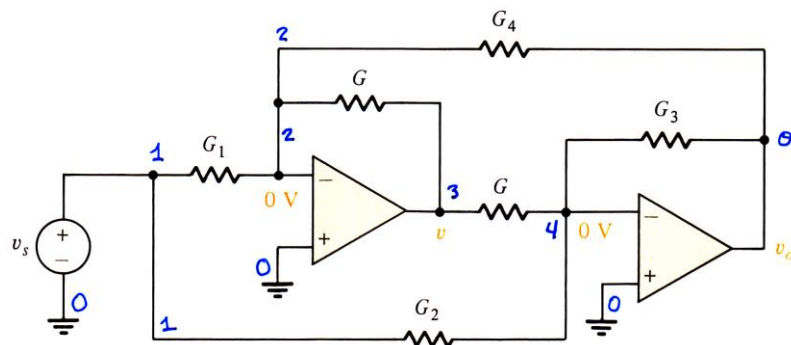


`true→s\si:s\dc("e1,1,0,8.:e2,2,0,8.01:o1,1,3,3:o2,2,4,4:r1,3,5,20k:r2,4,6,20k:r3,5,o,40k:r4,6,0,40k:o3,6,5,o:r5,o,0,10k"):ir5`

In the schematic, the current i_o corresponds to `ir5`. The answer, **$2.E-6$** , is correct.

Bo2's Example 3.3 (Cascade)

Find v_o in terms of the conductances and the applied voltage v_s .



`s\dc("e,1,0,vs:r12,1,2,1/g1:r14,1,4,1/g2:r4o,4,o,1/g3:r2o,2,o,1/g4:r23,2,3,1/g:r34,3,4,1/g:o1,0,2,3:o2,0,4,o")`

Evaluating `vo` we get:

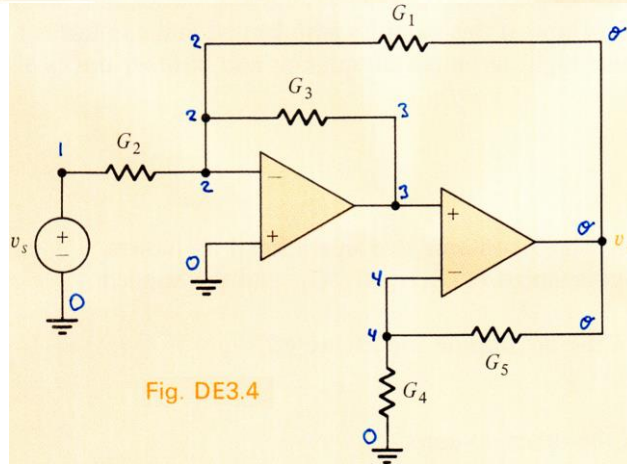
$$\frac{(g_1 - g_2) v_s}{g_3 - g_4}$$

which is correct, as can be seen by comparing it to the book's answer, shown below:

$$v_o = \frac{G_1 - G_2}{G_3 - G_4} v_s$$

Bo2's Drill Exercise 3.4 (Cascade)

For the op-amp circuit shown in Fig. DE3.4, suppose that $G_1 = 1 \text{ S}$, $G_2 = 2 \text{ S}$, $G_3 = 3 \text{ S}$, $G_4 = 4 \text{ S}$, and $G_5 = 5 \text{ S}$. Find v_o in terms of v_s . Answer: $-0.75v_s$

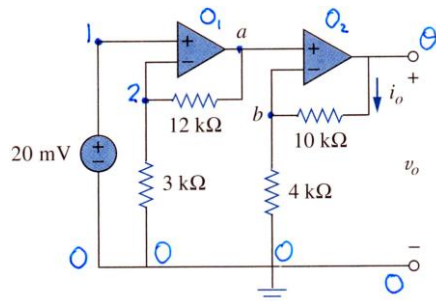


`s\dc("e,1,0,vs:r1,2,o,1/1:r2,1,2,1/2:r3,2,3,1/3:r4,4,0,1/4:r5,4,o,1/5:o1,0,2,3:o2,3,4,o")`

Evaluating v_o we get: **$-0.75 v_s$** , which is correct.

AS2's Example 5.9 (Cascade)

Find v_o and i_o .

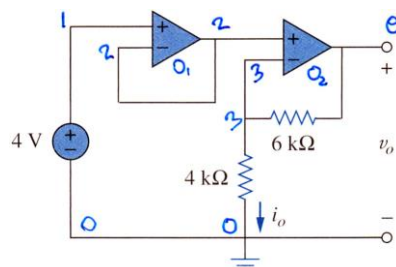


`true→s\si:s\dc("e,1,0,20m:o1,1,2,a:o2,a,b,o:ro,o,b,10k:r4,b,0,4k:r2,a,2,12k:r3,2,0,3k")`

Evaluating `approx({v_o,i_o})` we get the answer, **$\{0.35, 2.5 \times 10^{-5}\}$** . This is correct.

AS2's Practice Problem 5.9 (Cascade)

Determine v_o and i_o .

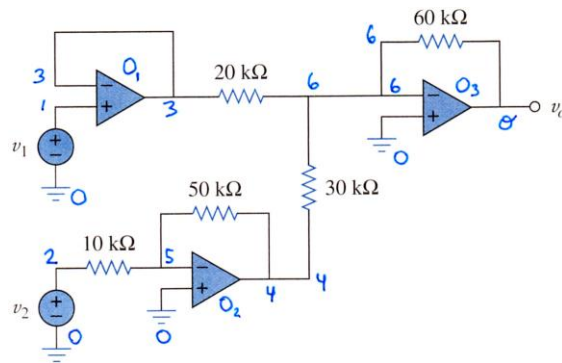


`true→s\si:s\dc("e,1,0,4:o1,1,2,2:o2,2,3,o:ro,3,0,4k:r6,3,0,6k"):{v_o,i_o}`

The answer, **{10,1/1000}**, is correct. To say a 1/1000 A current is the same as 1mA.

AS2's Practice Problem 5.10 (Cascade)

If $v_1 = 2\text{V}$ and $v_2 = 1.5\text{V}$, find v_o in the circuit below.

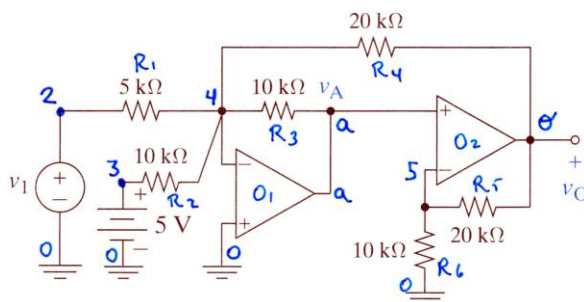


```
true→s\si:s\dc("e1,1,0,2:e2,2,0,1.5:o1,1,3,3:r1,2,5,10k:r2,3,6,20k:r5,5,4,50k:o2,0,5,4:r3,6,4,30k:r6,6,o,60k:o3,0,6,o"):vo
```

The answer, **9.**, is correct.

TR5's Example 4-16 (Cascade)

Derive an expression for v_o in terms of the two inputs.

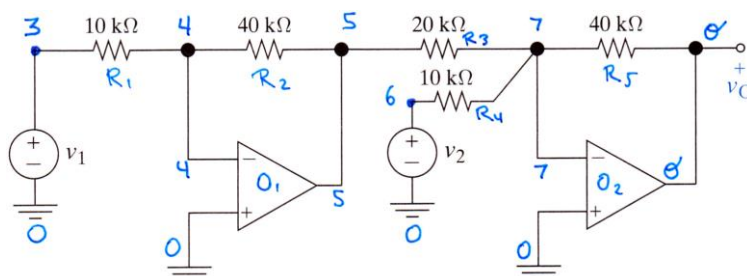


```
true→s\si:s\dc("e1,2,0,v1:e5,3,0,5:r1,2,4,5k:r2,3,4,10k:o1,0,4,a:r3,4,a,10k:r4,4,o,20k:o2,a,5,o:r5,5,o,20k:r6,5,0,10k"):expand(approx(vo))
```

The answer, **$-2.4 v_1 - 6$** , is correct.

TR5's Exercise 4-14 (Cascade)

Derive an expression for v_o in terms of the inputs v_1 and v_2 .



```

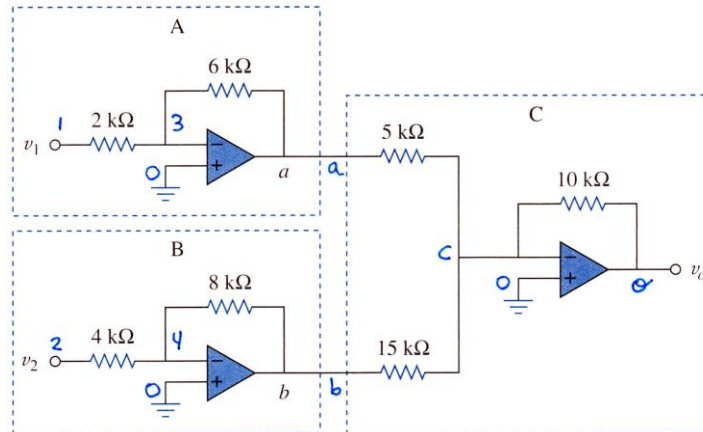
true→s\si:s\dc("e1,3,0,v1:r1,3,4,10k:o1,0,4,5:r2,4,5,40k:r3,5,7,20k:
e2,6,0,v2:r4,6,7,10k:r5,7,0,40k:o2,0,7,o"):expand(vo)

```

The answer, **8 v1 - 4 v2**, is correct.

AS2's Example 5.10 (Cascade)

If $v_1 = 1V$ and $v_2 = 2V$, find v_o in the circuit below.



```

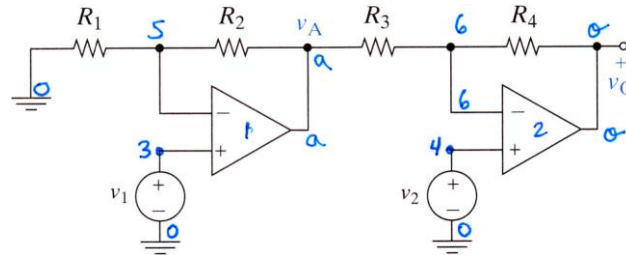
true→s\si:s\dc("e1,1,0,1:e2,2,0,2:r2,1,3,2k:r4,2,4,4k:r6,3,a,6k:r8,4,b,8k:r5,a,c,5k:r15,b,c,1
5k:r10,c,o,10k:oa,0,3,a:ob,0,4,b:oc,0,c,o"):approx(vo)

```

The answer, **8.667**, is correct.

TR5's Example 4-17 (Cascade)

Derive an expression for v_o in terms of the inputs v_1 and v_2 .



```

s\dc("r1,5,0,r1:r2,5,a,r2:r3,a,6,r3:r4,6,o,r4:e1,3,0,v1:e2,4,0,v2:o1,3,5,a:o2,4,6,o"):vo

```

We get the right answer. Let's compare it with the book's answer, which is below:

$$-\left[\frac{R_4}{R_3}\right]\left[\frac{R_1 + R_2}{R_1}\right]v_1 + \left[\frac{R_3 + R_4}{R_3}\right]v_2$$

Our answer, expanded via `expand(vo)`, is shown below:

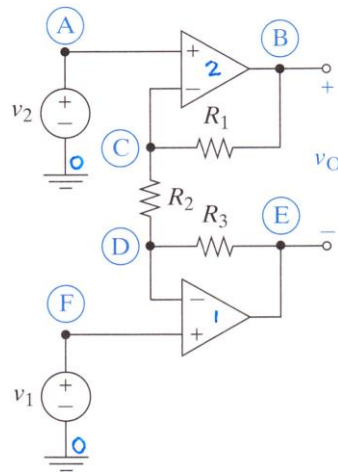
$$\frac{-r2 r4 v1}{r1 r3} - \frac{r4 v1}{r3} + \frac{r4 v2}{r3} + v2$$

Playing with it by hand we get a form that, in my opinion, is *prettier* ;-) than the book's:

$$v2 \left(\frac{r4}{r3} + 1 \right) - v1 \left(\frac{r4}{r3} \right) \left(\frac{r2}{r1} + 1 \right)$$

TR5's Example 4-18 (Multiple)

Derive an expression for v_o in terms of the inputs v_1 and v_2 .



`s\dc("e2,a,0,v2:e1,f,0,v1:o2,a,c,b:o1,f,d,e:r1,b,c,r1:r2,c,d,r2:r3,d,e,r3")`

To find v_o , we ask for **vb-ve**. We get an expression that can easily be rearranged to look like this:

$$-\left(\frac{r1 + r2 + r3}{r2}\right)(v1 - v2)$$

This is exactly the answer from the book:

$$v_o = v_B - v_E = \left[\frac{R_1 + R_2 + R_3}{R_2} \right] (v_2 - v_1)$$

If you are ever in doubt about whether two expressions are the same, enter them both, separately, into the calculator, and then ask the calculator to compare them using the equality sign. If the answer is **'true'**, then they are the same.

That is it for direct current (DC) analysis of linear circuits using Symbulator, and marks the end of Part 1.